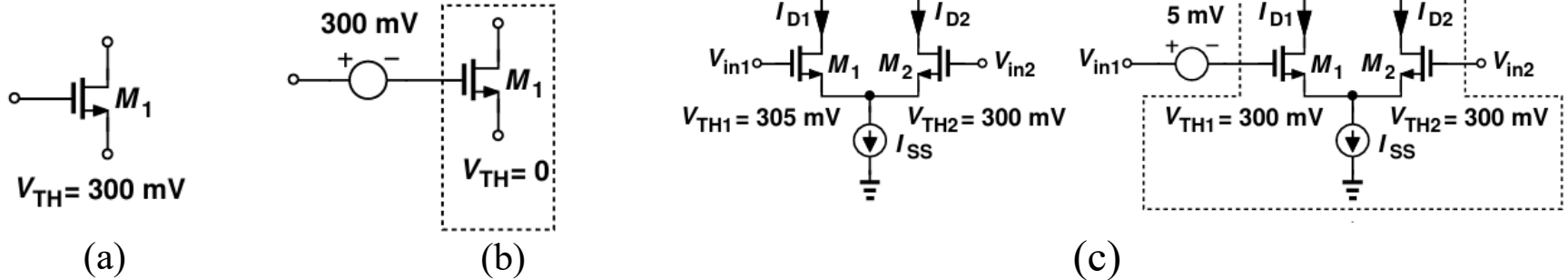


Comparators

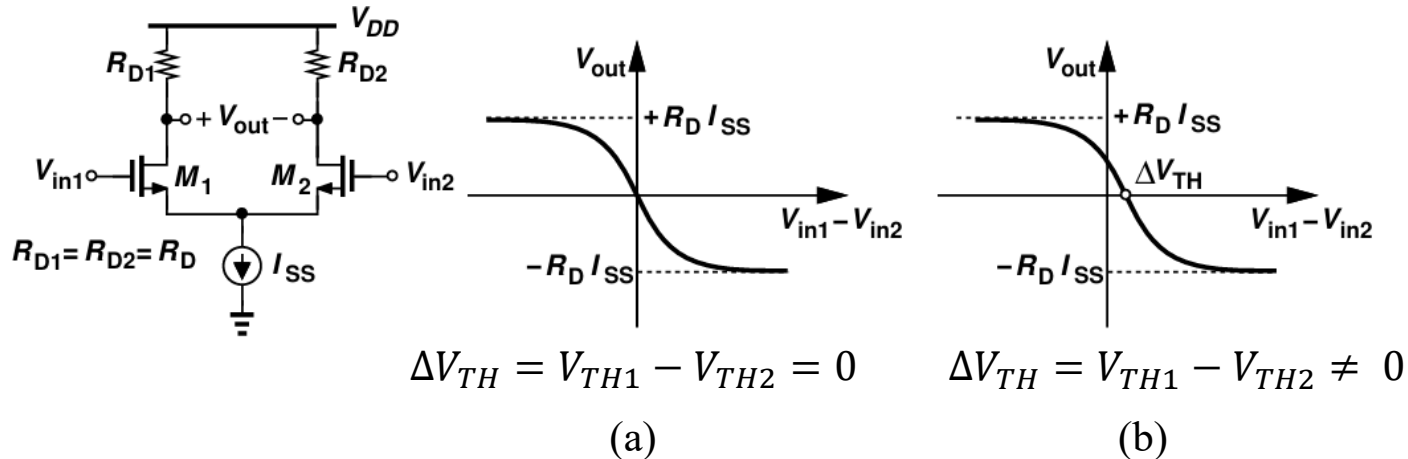
- **Input referred offset for comparator**
- **Amplification by positive feedback**
- **Current steering comparator**
- **Comparator parameters**
 - **Input offset, Speed, Noise, Kickback**
- **Differential Pairs with Capacitive Loads**
- **Addition of positive feedback to differential pair with capacitive loads**
- **Complete StrongArm Comparator**
- **Properties of StrongArm Comparator**

Offset



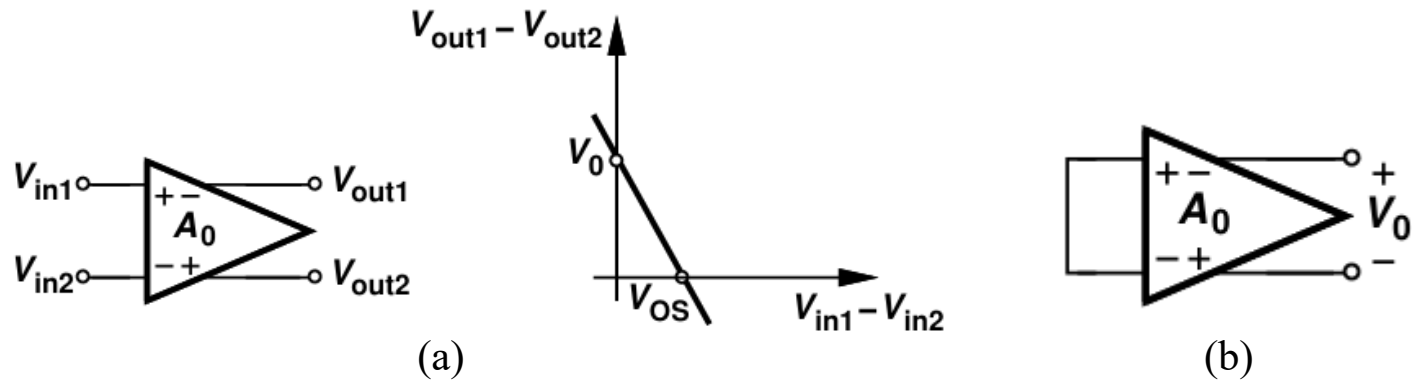
- Figures (a) and (b) are identical since MOSFET equations are always functions of $V_{GS} - V_{TH}$.
- From the above argument, the two configurations in (c) are identical.

Input referred offset



- In figure (b), V_{out} vs $V_{in1} - V_{in2}$ characteristics shift right by ΔV_{TH} amount because of the Threshold voltage mismatch between M_1 and M_2 .
- The input differential voltage for which output voltage goes to zero is called “input referred offset voltage”.
- ΔV_{TH} is the input referred offset voltage (V_{OS}) for this circuit.

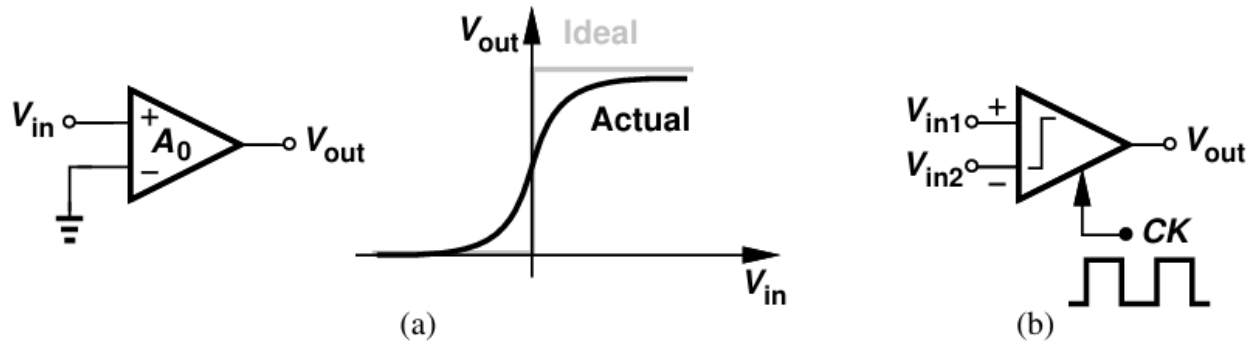
Input offset calculation



- Calculating input offset for complex cases:
 - a) Set input to 0, measure V_{out} (V_0), shown in Fig(b).
 - b) Calculate small signal gain (A_0).

$$V_{OS} = \frac{V_0}{A_0}$$

Clocked comparator



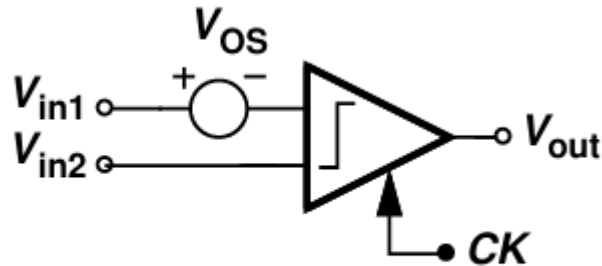
- Objective of comparator is to generate logical output high or low.

$V_{out} = \text{Low if } V_{in} < 0$

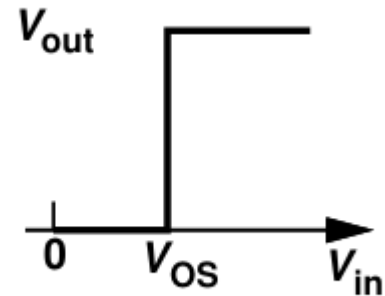
$= \text{High if } V_{in} > 0$

- Clocked comparator makes a decision when CK goes high.

Comparator offset



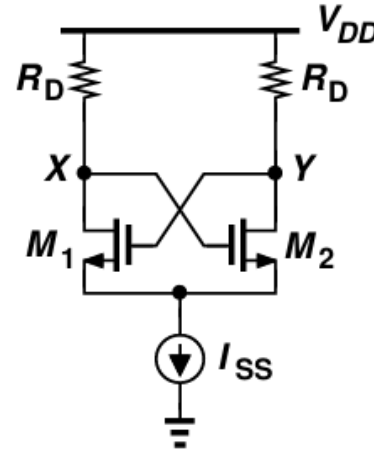
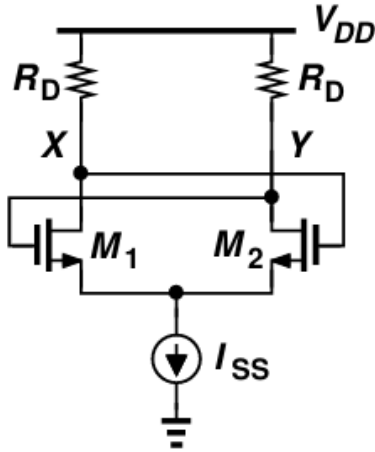
(a)



(b)

- V_{OS} is the input referred offset due to all asymmetries in the comparator.
- If the input difference $V_{in1} - V_{in2} < V_{OS}$ then comparator gives incorrect decision.
- In a typical ADC design, V_{OS} is in the range of 0.1 to 0.2 LSB.

Amplification by positive feedback



- An ideal comparator has a very large gain at the threshold point. Realizing comparator with Op-amp consumes high power (Gain bandwidth tradeoff).
- This tradeoff is overcome by using amplification by positive feedback.

Frequency domain perspective

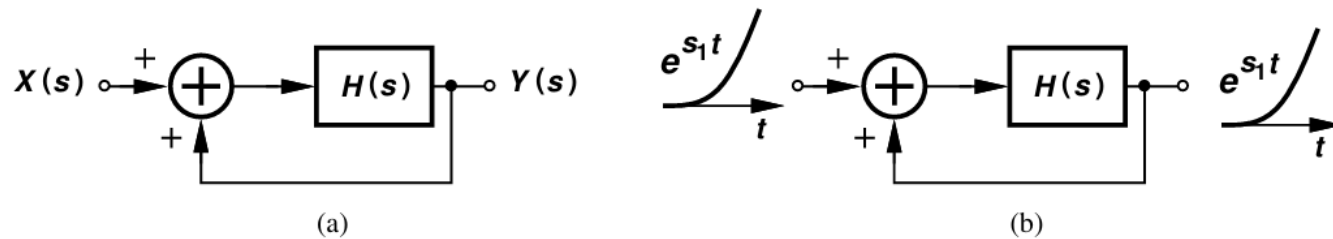


Figure 5.10 (a) System with positive feedback, and (b) growing exponential circulating around the loop.

- Input $e^{s_1 t}$ to a LTI system of transfer function $H(s)$ produces $H(s_1) e^{s_1 t}$.
- For feedback loop in Fig 5.10 (a)

$$\frac{Y}{X} = \frac{H(s)}{1 - H(s)}$$

- For $H(s) = \frac{A_0}{1 + \frac{s}{w_0}}$, $\frac{Y}{X} \rightarrow \infty$ when $s_1 = (A_0 - 1)w_0$.

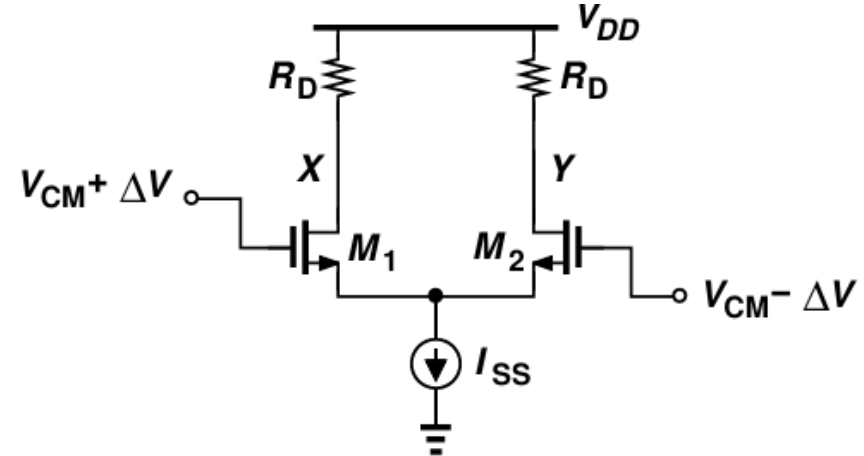
- For differential amplifier in Fig 5.9, $A_0 = g_m R_D$, $w_0 = \frac{1}{R_D C_L}$

$$s_1 = \frac{g_m R_D - 1}{R_D C_L}$$

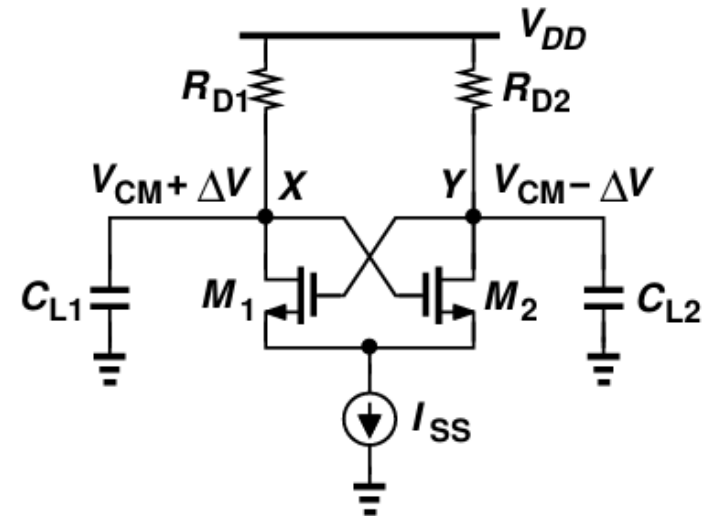
- Thus the loop in Fig 5.9 has exponential growth if $g_m R_D > 1$ (i. e, $s_1 > 0$)

Time domain perspective

- $V_X - V_Y = -2g_m R_D \Delta V$, for input difference of $2\Delta V$.



- If the inputs and outputs are tied in cross coupled fashion, an imbalance of $2\Delta V$ at gate leads to $2g_m R_D \Delta V$ at drain.
- Since drains and gates are shorted, this further leads to $2(g_m R_D)^2 \Delta V$ and so on, which results in exponential growth if $g_m R_D > 1$.



Small signal time domain perspective

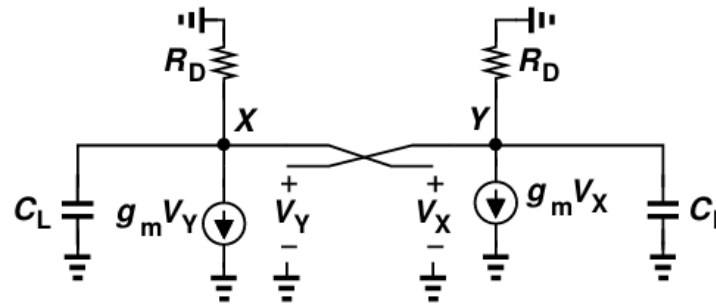
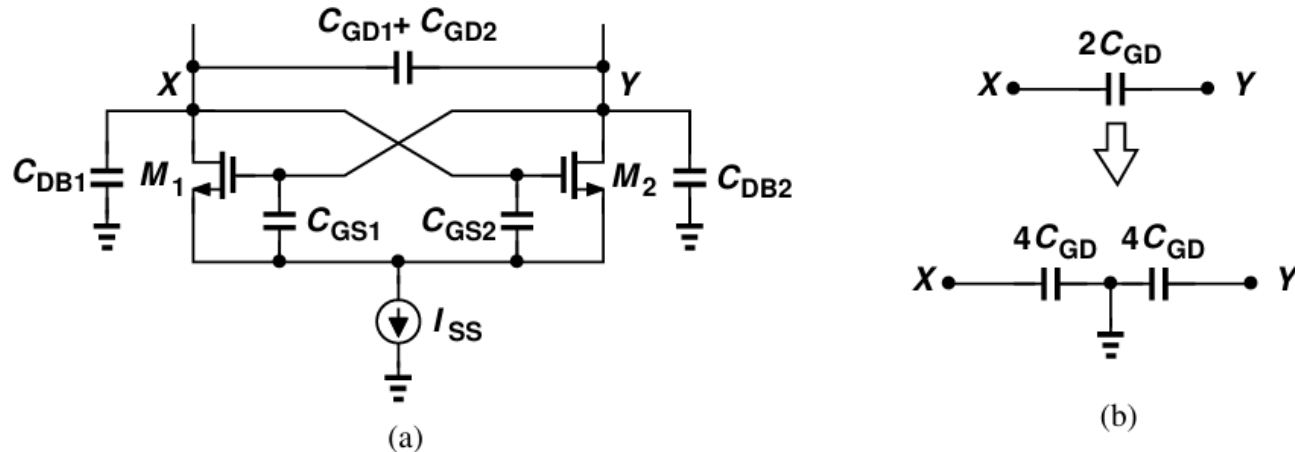


Figure 5.15 Small-signal model of cross-coupled pair.

$$\begin{aligned}
 C_L \frac{dV_X}{dt} + \frac{V_X}{R_D} + g_m V_Y &= 0 \\
 C_L \frac{dV_Y}{dt} + \frac{V_Y}{R_D} + g_m V_X &= 0 \\
 \Rightarrow V_{XY} &= V_{XY0} \exp \frac{t}{\frac{R_D}{g_m R_D - 1} C_L}
 \end{aligned}$$

Regeneration time constant, $\tau_{reg} = \frac{R_D}{g_m R_D - 1} C_L$

Parasitic capacitance in cross coupled pair

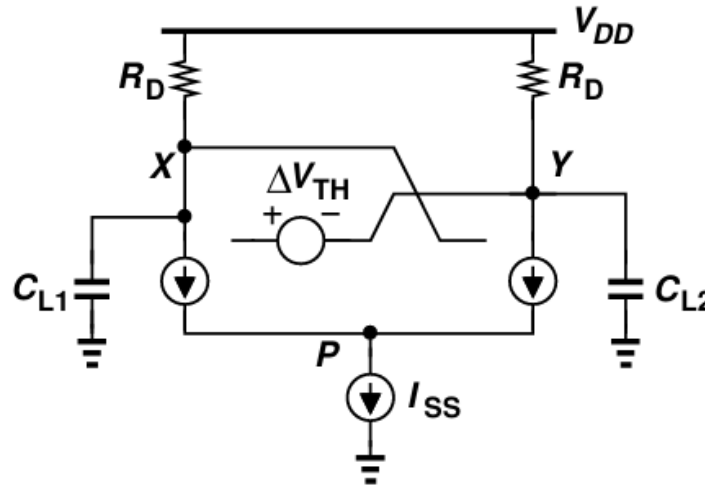


- Even if there is no explicit load to the cross coupled pair, the parasitic capacitances of MOSFETs acts as C_L .
- In the above configuration,

$$C_{L1} = C_{L2} = C_{GS} + C_{DB} + 4C_{GD}$$

- If R_D is very large, regeneration time constant = $\frac{C_L}{g_m}$.
- So, the inherent speed of the circuit is proportional to f_T of the transistor.

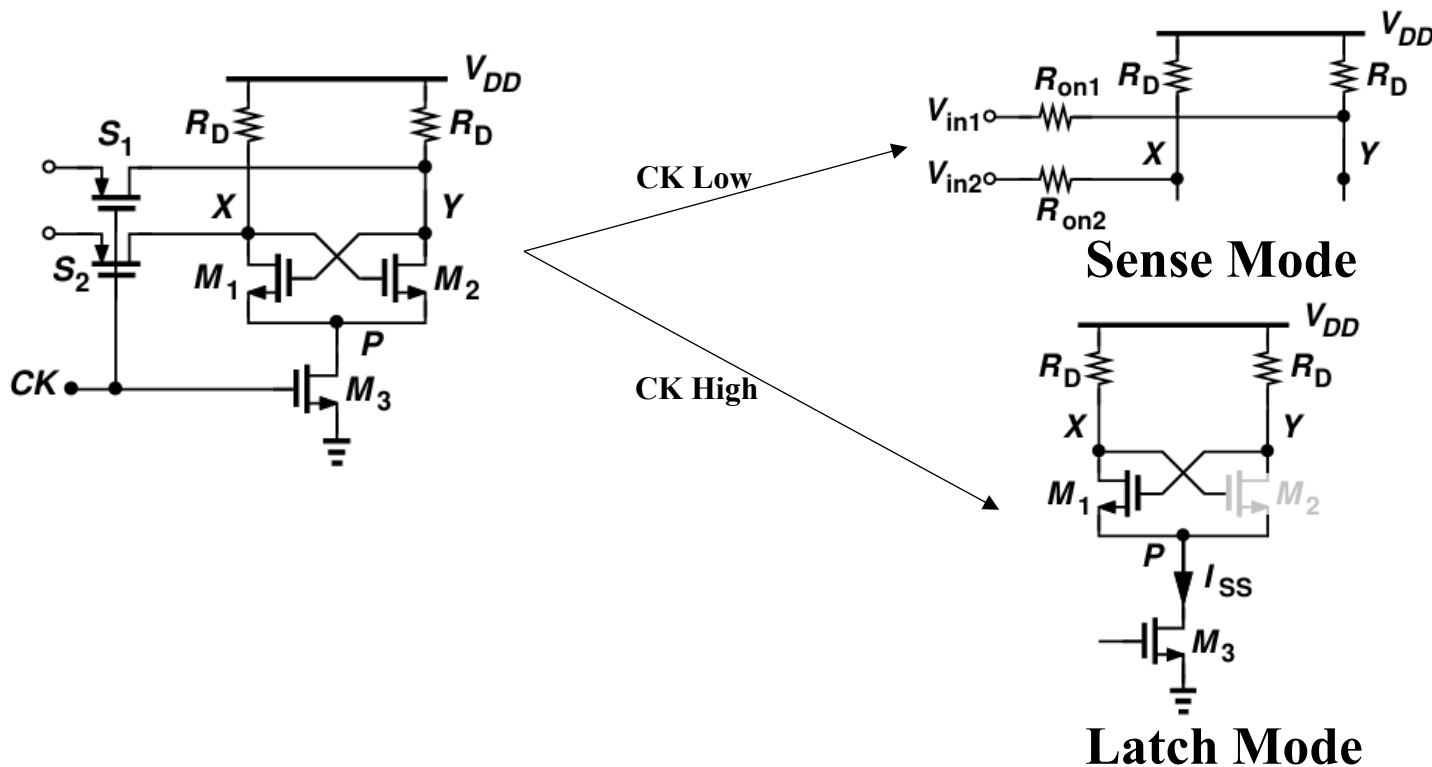
Uncontrolled positive feedback



- If there is threshold mismatch (ΔV_{TH}) between transistors of cross coupled pair, the circuit amplifies its own offset and makes a logical decision even in the absence of imbalance between V_X and V_Y initially.
- Input referred offset for this circuit is,

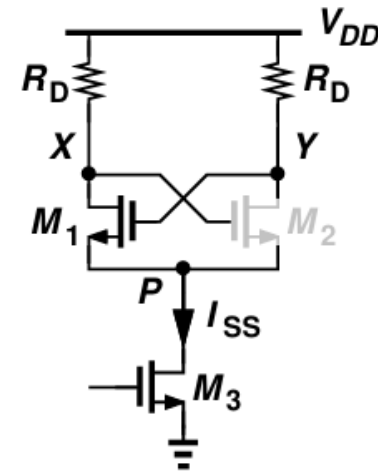
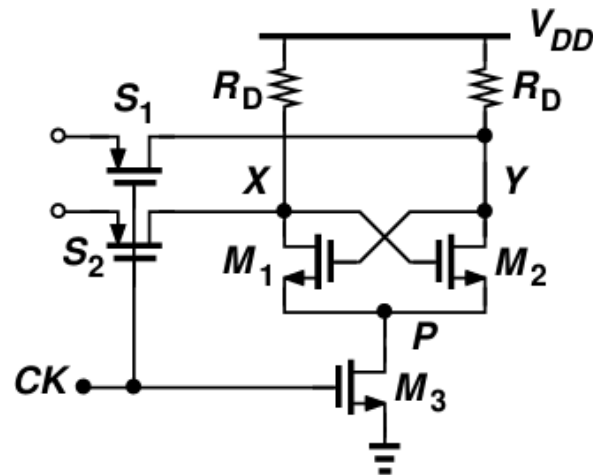
$$V_{OS} = \frac{g_m R_D}{g_m R_D - 1} \Delta V_{TH}$$

Simple comparator



- In “Sense mode”, CK is low and the input difference is stored on X, Y nodes.
- In the “Latch mode”, inputs are disconnected and the difference stored on X, Y nodes is amplified to V_{DD} and $V_{DD} - R_D I_{SS}$.

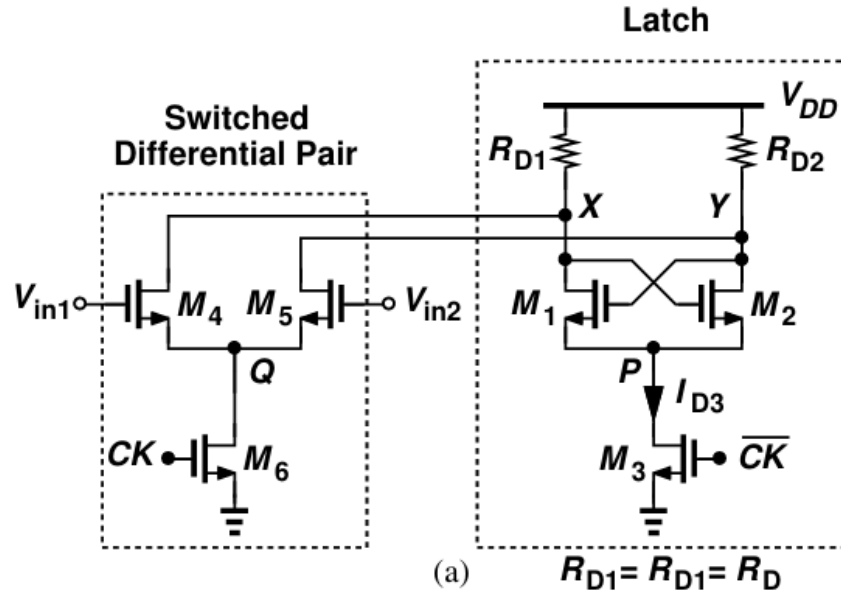
Drawbacks of simple comparator



Latch Mode

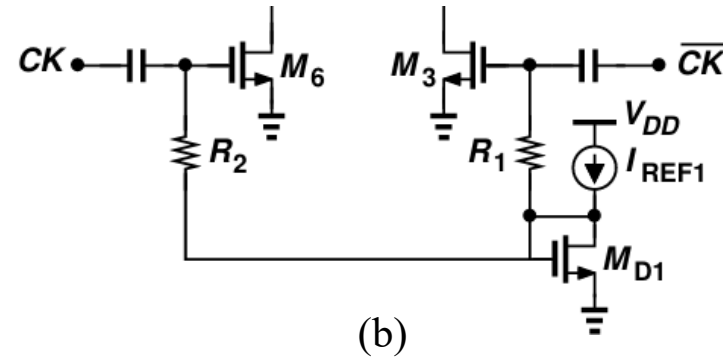
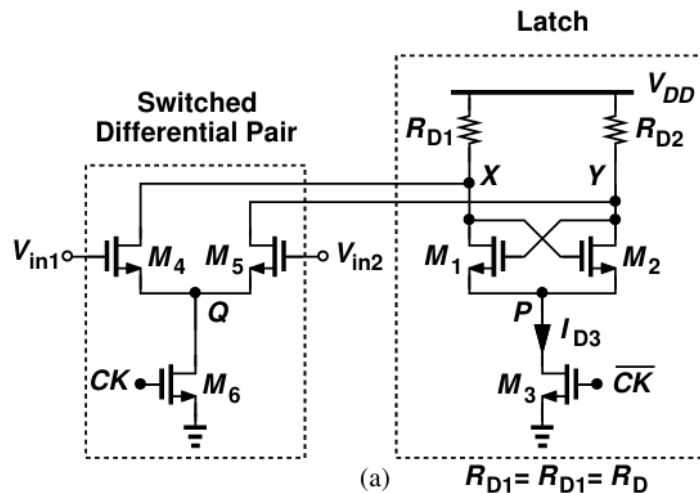
- Bias current of M_3 is poorly defined in Latch mode.
- In Latch mode, $V_P = V_{DD} - R_D I_{SS} - V_{DS1}$ while gate of M_3 is at V_{DD} . This pushes M_3 to Triode.
- There is no voltage gain before regeneration begins.

Current steering comparator



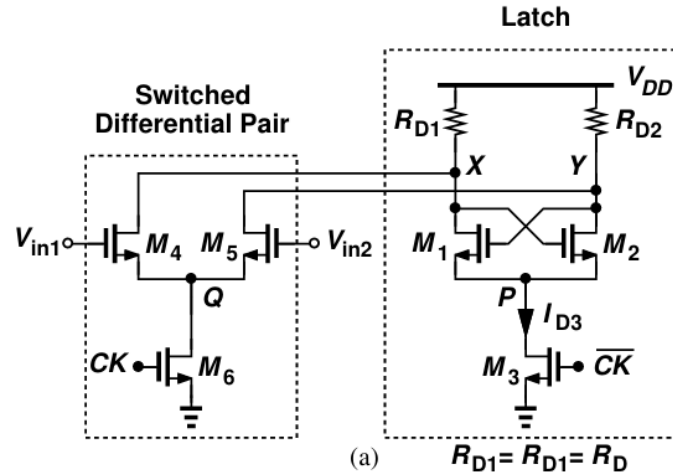
- In the “Sense mode” CK is high, input difference is amplified by $g_{m4,5}R_D$ and stored on X, Y nodes.
- In the “Latch mode” CK is low, difference between X, Y nodes is amplified.

Drawbacks of current steering comparator



- Bias currents of M_3 and M_6 lead to static power consumption.
- Output swings are not rail-to-rail, requiring further amplification to drive CMOS logic.
- Coupling capacitors occupy large area, which is a big issue if a large number of comparators are required for ADC.

Comparator parameters –Input offset



- In the “sense mode”, input referred offset is given by

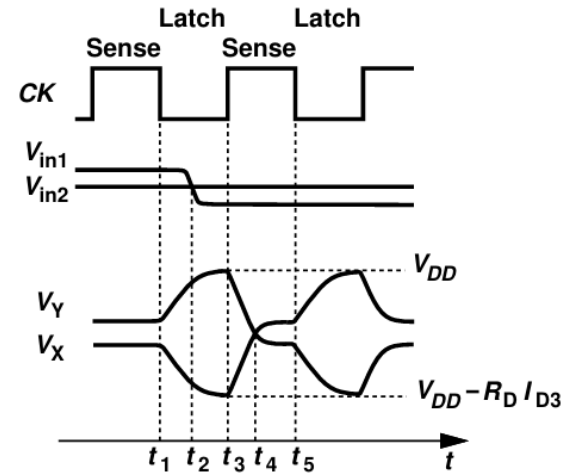
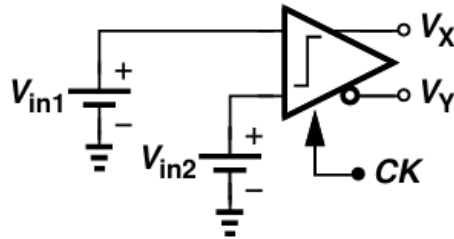
$$V_{OS, sen}^2 = \Delta V_{TH4}^2 + \left(\frac{I_{SS}}{2 g_{m4,5} R_D} \right)^2 * \Delta R_D^2$$

- In the “latch mode”, mismatches between M_1 and M_2 add to input referred offset.

$$V_{OS, sen}^2 = \Delta V_{TH4}^2 + \left(\frac{I_{SS}}{2 g_{m4,5} R_D} \right)^2 \Delta R_D^2 + \left(\frac{g_{m1,2}}{g_{m4,5} (g_{m1,2} R_D - 1)} \Delta V_{TH} \right)^2$$

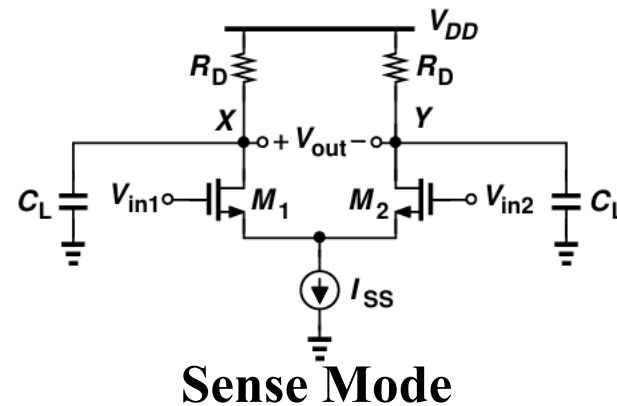
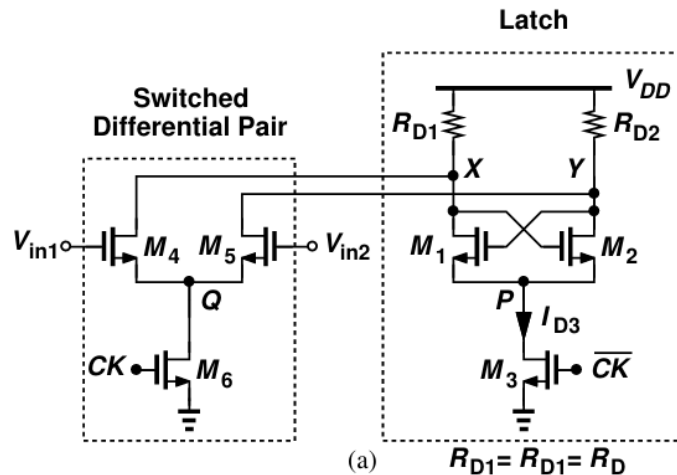
- Imbalance due to resistors in latch mode need not be added additionally, since they are already accounted for in sense mode.

Comparator parameters – Speed



- The comparator is said to operate at max. speed of f_{CK} if the outputs get resolved to V_{DD} and $V_{DD} - R_D I_{SS}$ at the end of latch mode correctly.
- In a typical test input difference of 0.5LSB to check the speed of comparator.
- For a more severe test, a small positive value is applied in one cycle and large negative in another.
- Since the input to comparator random, very small input voltages lead to “metastability”.

Comparator parameters – Noise



- In “sense mode”,

$$V_{n,out}^2 = 2 (4kTR_D + 4kT \gamma g_{\{m4,5\}} R_D^2)$$

- This noise gets filtered by low pass filter with pole at $\frac{1}{R_D C_L}$,

$$V_{n,tot,sen}^2 = \frac{2(1+\gamma g_{m4,5} R_D)kT}{C_L}$$

- In “latch mode”, input referred noise can be calculated similar to input referred offset

$$V_{n,tot,latch}^2 = \left(\frac{4kT\gamma}{g_{m1,2}} \right) \frac{(g_{m1,2} R_D)^2}{(g_{m1,2} R_D - 1)^2}$$

Comparator parameters – Kickback noise

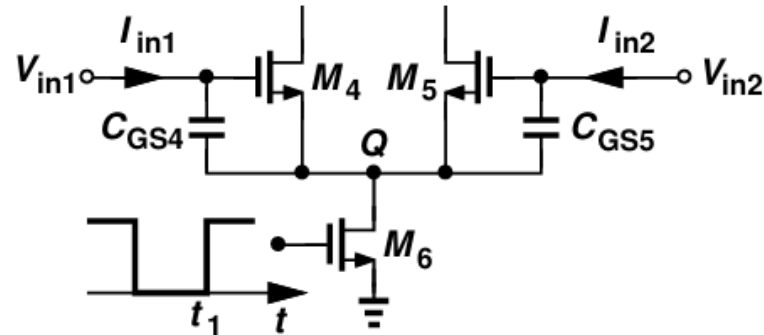
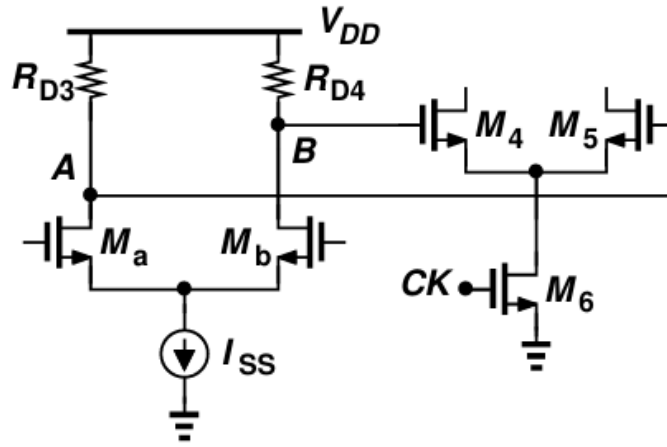


Figure 5.26 Kickback noise at input of switched pair.

- Switching action within a comparator cause transient currents to flow through its input terminals called as “kickback noise”.
- In above figure, when gate voltage of M_6 turns on, M_4 and M_5 are turned off. So, the current to discharge node Q comes from C_{GS4} and C_{GS5} which causes kickback.
- If these currents flow through unequal impedances, they introduce additional error in comparison.

Comparator parameters – Kickback noise



- To reduce kickback, comparator is preceded with pre amplifier (unswitched stage).
- In above figure when CK goes high, kickback from M_4 and M_5 must travel through C_{GDa} and C_{GDb} which gets suppressed.

Differential pair with capacitive load

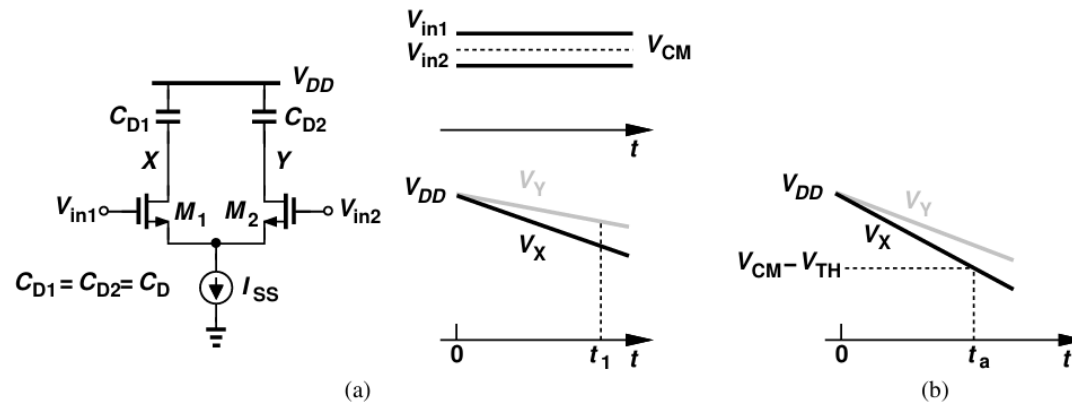


Figure 5.29 (a) Differential pair with capacitive loads, and (b) evolution of V_X and V_Y with time.

- Differential amplifier with resistive loads has gain $g_m r_0$. If the loads are capacitive, the gain is $\frac{g_m}{sC_L}$, which is very high at low frequencies.
- In time domain perspective as shown in above figure, V_X and V_Y start at V_{DD} . V_X falls faster than V_Y since $V_{in1} > V_{in2}$.
- Gain at given point in time, $\frac{V_X - V_Y}{V_{in1} - V_{in2}} = -\frac{g_m}{C_D} t$.
- Once M_1 and M_2 go to triode, g_m decreases. This circuit is considered amplifier till they are in saturation.

$$Gain = \left| \frac{V_X - V_Y}{V_{in1} - V_{in2}} \right| = 2 \frac{V_{DD} - (V_{CM} - V_{TH})}{V_{GS} - V_{TH}}$$

Differential pair with capacitive load

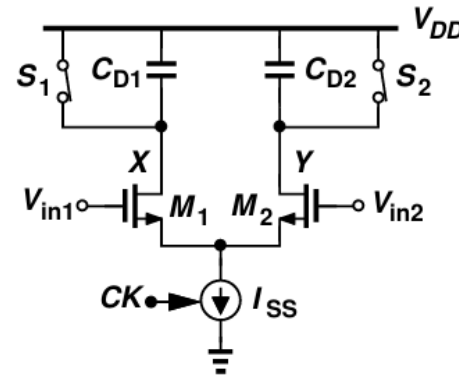


Figure 5.31 Addition of precharge.

- Once V_X and V_Y reaches zero, these nodes need to be precharged to V_{DD} before the next cycle begins.
- As shown in above diagram, switches S_1 and S_2 turn on making $V_X = V_Y = V_{DD}$.
- During this time CK is turned off to save power.

Addition of positive feedback to differential pair with capacitive loads

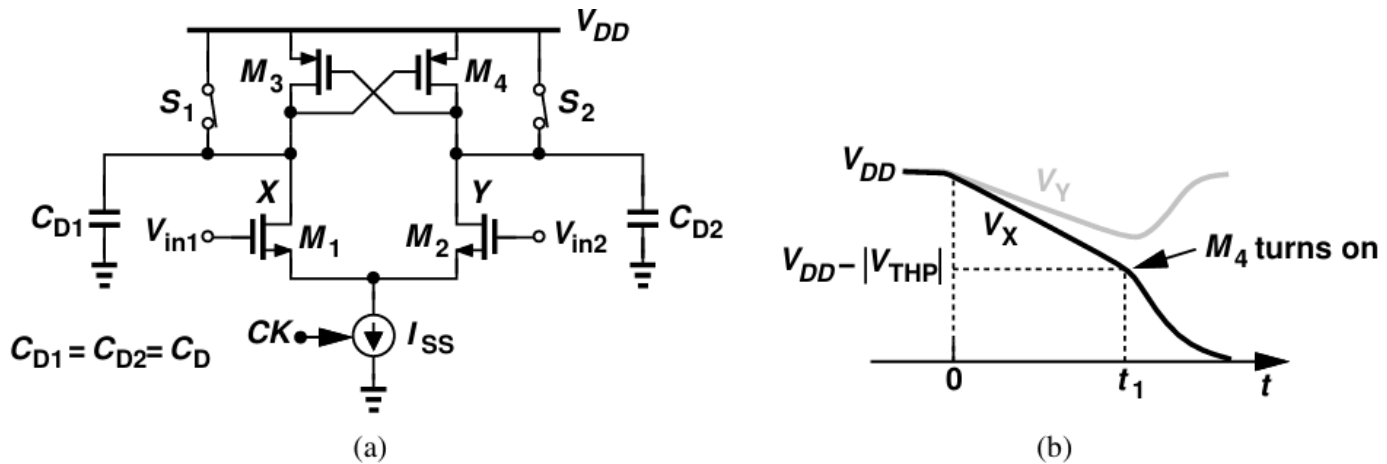


Figure 5.32 (a) Addition of positive feedback, and (b) corresponding waveforms.

- In “Precharge mode”, S_1 and S_2 are turned on and CK is off pulling V_X, V_Y to V_{DD} .
- In “Amplification mode”, S_1 and S_2 are turned off and CK is on.
- M_3, M_4 are off till “ t_1 ” i.e, $V_X, V_Y > V_{DD} - |V_{THP}|$.
- Once V_X reaches $V_{DD} - |V_{THP}|$, M_4 turns on opposing the fall and V_Y begins to rise.

Addition of positive feedback to differential pair with capacitive loads

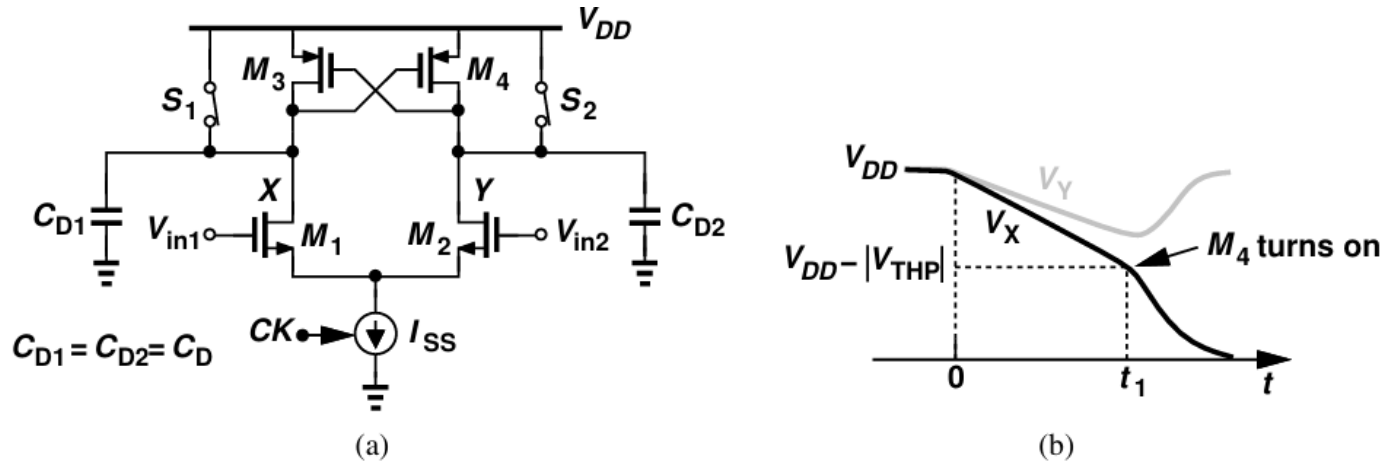


Figure 5.32 (a) Addition of positive feedback, and (b) corresponding waveforms.

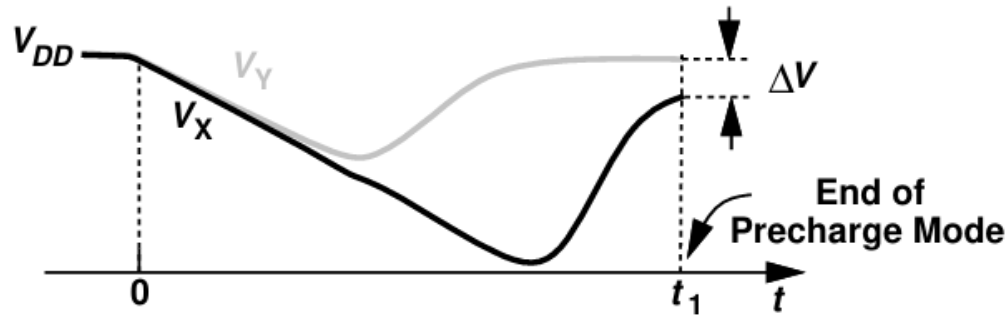
- Unlike previous topologies, “Sense” and “latch” functions occur while CK is on for this topology.
- Input difference is amplified till t_1 , suppressing the offsets due to M_3, M_4 .

$$\frac{V_X - V_Y}{V_{in1} - V_{in2}} (t = t_1) = -\frac{2g_{mN}}{I_{SS}} |V_{THP}|$$

- Overall input referred offset,

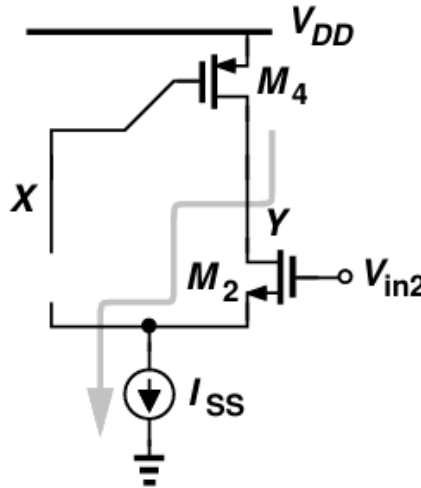
$$V_{OS}^2 = \Delta V_{TH1,2}^2 + \frac{\Delta V_{TH3,4}^2}{4g_{mN}^2 |V_{THP}|^2} I_{SS}^2$$

Dynamic offset



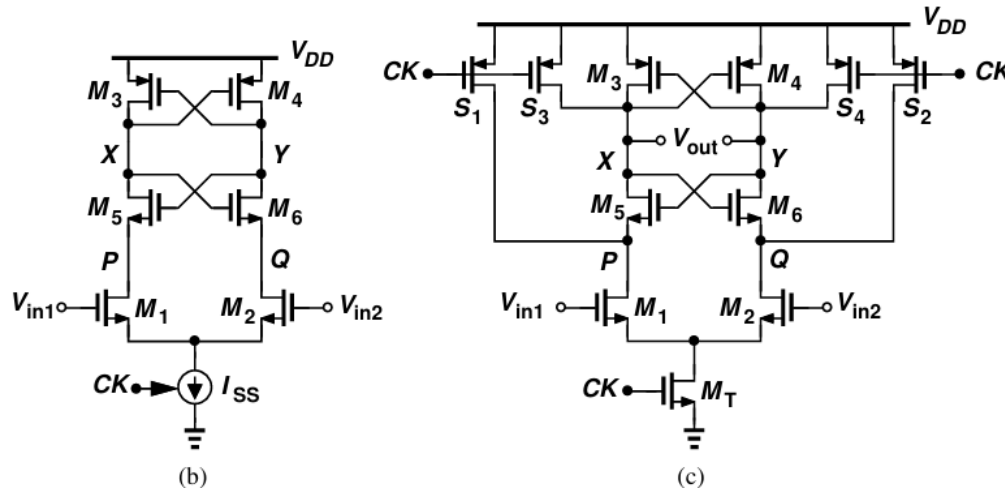
- At the end of precharge mode, difference between V_X, V_Y becomes offset in “amplification mode”.
- This dynamic offset error adds to $\Delta V_{TH3,4}$.
- In the previous slide, it is shown that offset is function of I_{SS} . If the input common mode changes with time, then so does the offset.

Drawbacks of positive feedback differential pair with capacitive loads



- This topology draws static current in amplification mode even after the logical decision made.
- Static current flow prohibits V_Y from reaching V_{DD} .

StrongArm Comparator

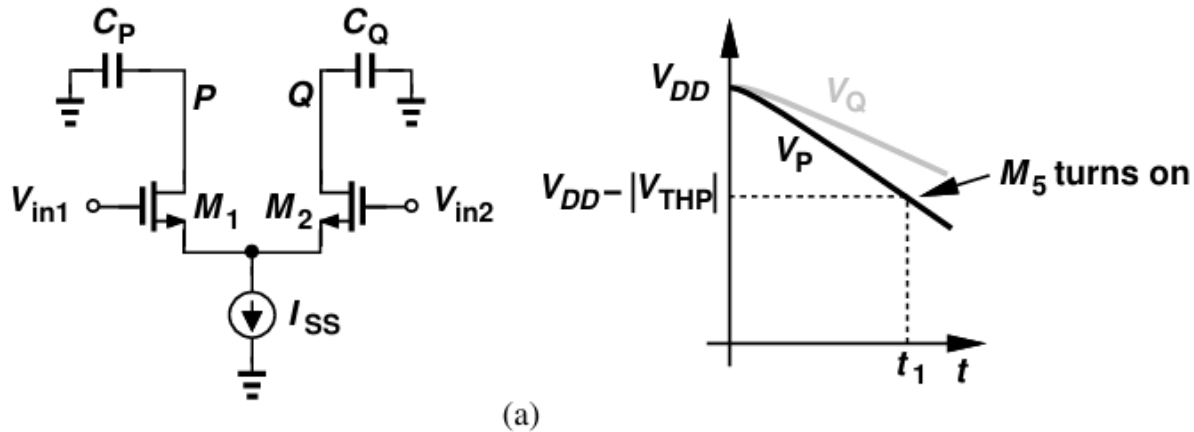


- NMOS M_5, M_6 added to eliminate static power consumption in “amplification mode”.

Mode 1:

- In precharge mode, M_T is turned off, switches $S_1 - S_4$ are turned on pulling the nodes V_X, V_Y, V_P, V_Q to V_{DD} .

StrongArm Comparator



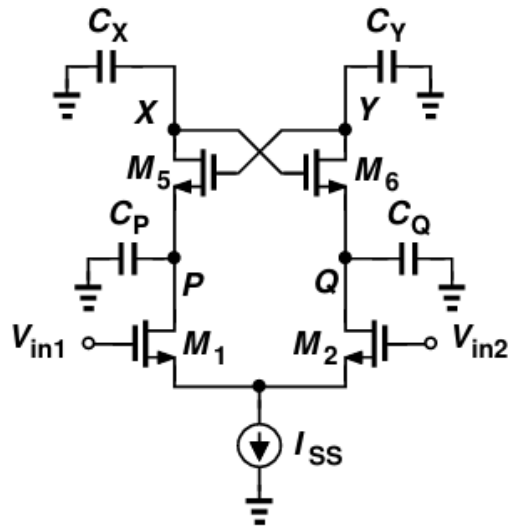
Mode 2:

- M_1, M_2 are in saturation and V_P falls faster than V_Q given $V_{in1} > V_{in2}$.
- This mode ends when either of V_P, V_Q falls to $V_{DD} - |V_{THP}|$, turning on M_5 or M_6 .

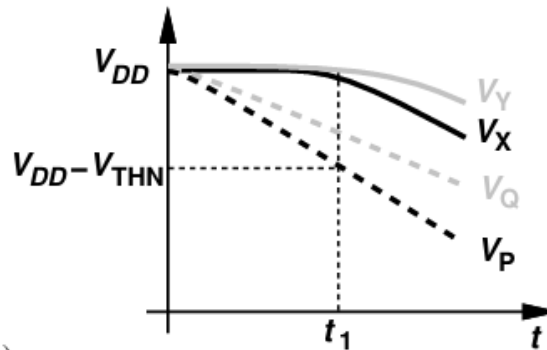
$$\frac{V_P - V_Q}{V_{in1} - V_{in2}} (t = t_1) = -\frac{2g_{m1,2}}{I_{SS}} V_{THN}$$

- This gain helps reduce offset contributed by M_5, M_6 .

StrongArm Comparator



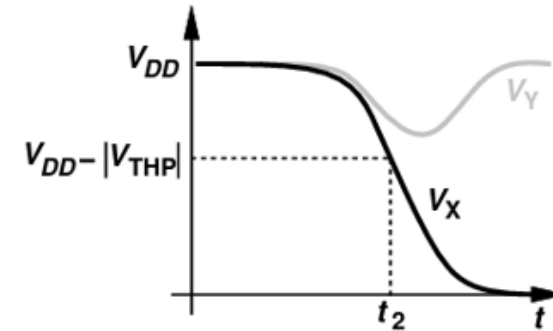
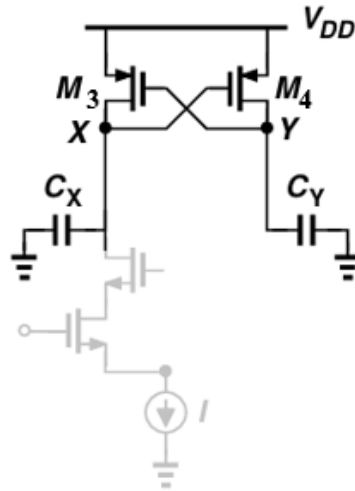
(b)



Mode 3:

- M_5, M_6 are turned on, V_X, V_Y begin to fall.
- Two amplification mechanisms
 - a) I_{D1}, I_{D2} flow through C_X, C_Y .
 - b) M_5, M_6 provide regeneration (which is weak)
- This mode ends when V_X, V_Y falls to $V_{DD} - |V_{THP}|$, turning on M_3, M_4 .

StrongArm Comparator



(c)

Mode 4:

- Positive feedback action of cross coupled PMOS starts, pulling one of the node to V_{DD} and the other node goes to 0.
- The regeneration equation is,

$$V_{XY} = V_{XY0} e^{\left(\frac{t}{\frac{r_{O3,4}}{g_{m3,4}r_{O3,4}-1} C_{X,Y}} \right)}$$

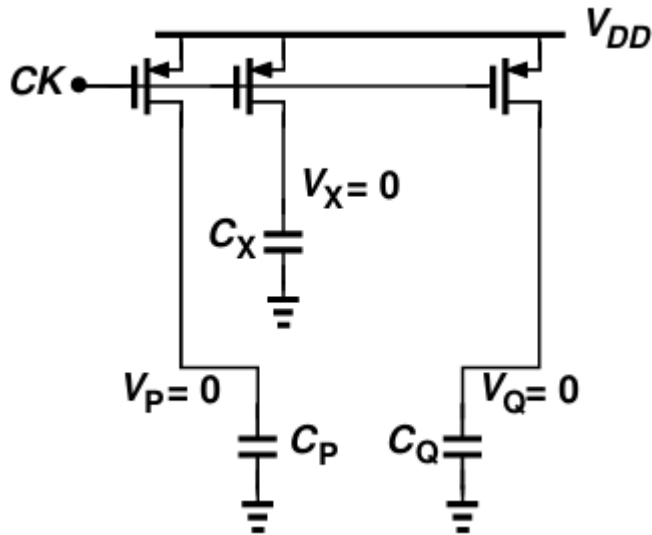
- $V_{X,Y0}$ is difference between V_X, V_Y when M_3, M_4 turns on.

Advantages of StrongArm comparator

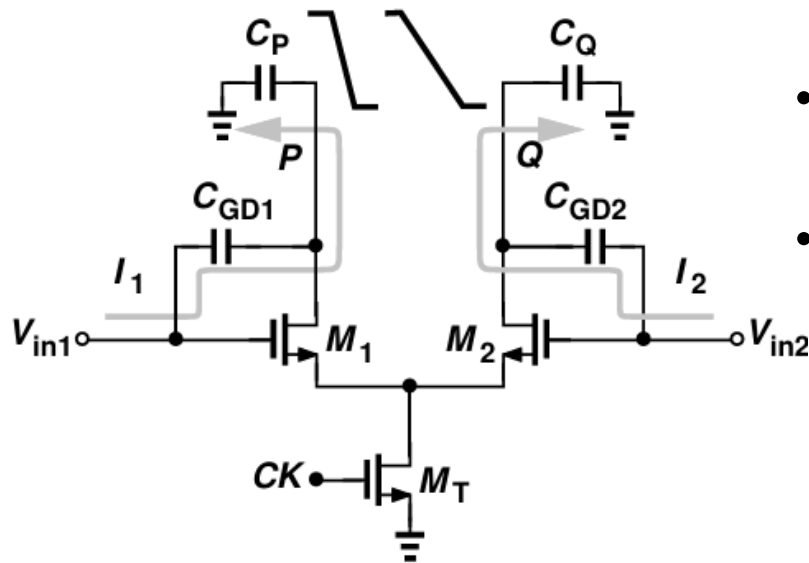
- Delivers rail to rail outputs
- No static power consumption.
- Contribution to V_{OS} by $M_3 - M_6$ is low due to initial voltage gain provided in mode 2.
- Comparator requires only single clock phase.
- Low average power consumption.

$$P \approx f_{CK}(C_X + C_P + C_Q)V_{DD}^2$$

Cons of StrongArm comparator



- At the start of precharge mode, $S_1 - S_4$ start in saturation and draw large transient currents from power supply momentarily.



- Strong arm topology has high kickback noise.
- In addition to kick back noise when M_T turns on, V_P and V_Q fall at unequal rates and this gets coupled to inputs through C_{GD1} , C_{GD2} .

Need for RS Latch

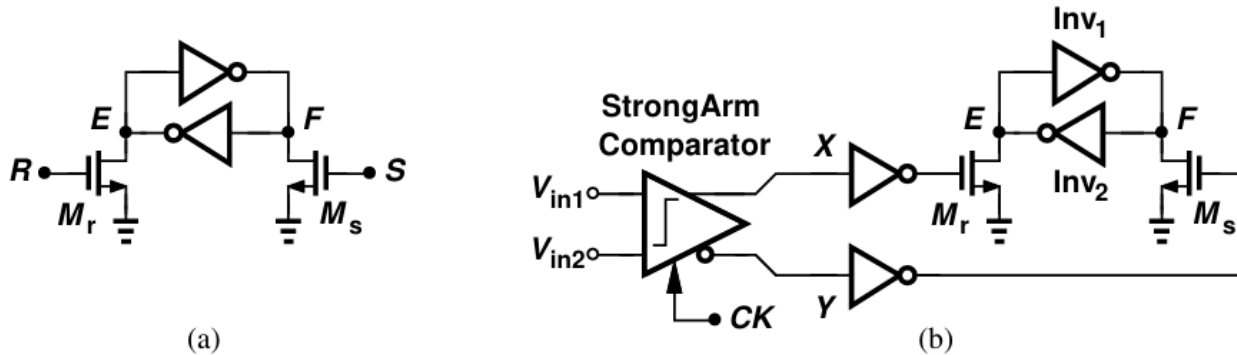


Figure 5.43 (a) An RS latch, and (b) its interface with StrongArm comparator.

- In precharge mode, both outputs of comparator are high which don't work well with the subsequent digital logic.
- Additional latch is added to retain previous logic levels.
- When CK is low in precharge mode, V_E , V_F retain previous logic levels.
- When CK is high, V_X , V_Y produces differential voltages driving nodes V_E , V_F to differential values.
- M_r , M_s should be strong enough to overcome PMOS in Inv_2 and Inv_1 respectively.

Basic Concepts

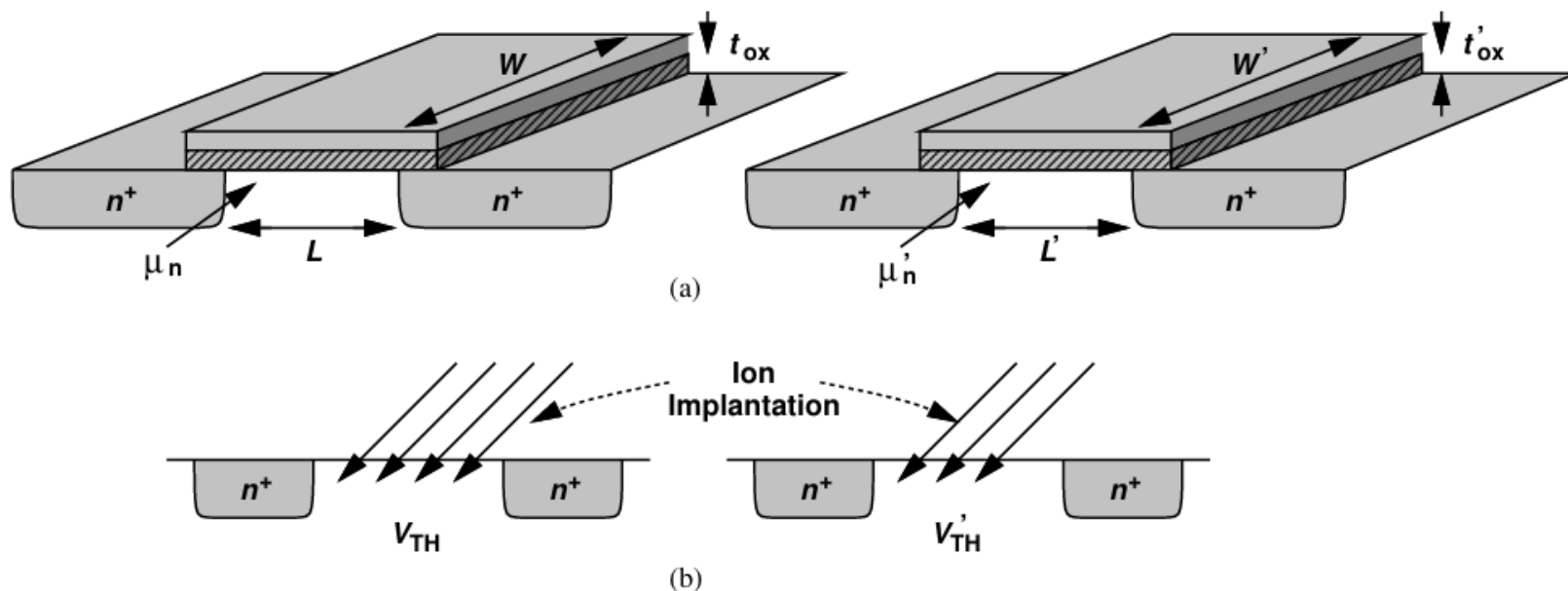
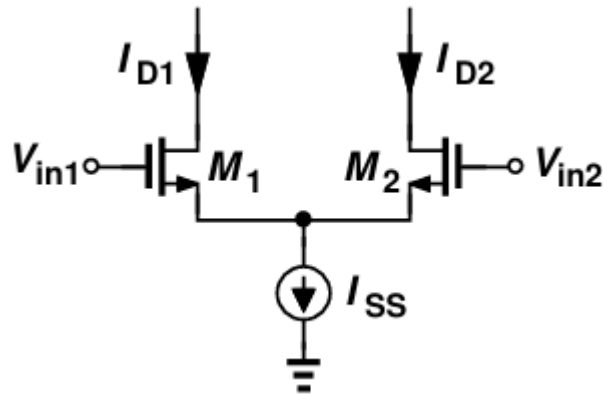


Figure 6.1 (a) Mismatch between two transistors, and (b) effect of implant mismatch .

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})^2, \quad \sigma_{tot} = \sqrt{\sigma_1^2 + \sigma_2^2}.$$

- **W, L & t_{ox} mismatch**
- **Maximizing W & L minimizes mismatch**
- **Slight V_{TH} mismatch**

Example: Differential pair



$$V_{OS,in}^2 = \frac{1}{4} \left(\frac{\Delta\beta}{\beta} \right)^2 (V_{GS} - V_{TH})^2 + \Delta V_{TH}^2.$$

- 2 choices: wide transistors or small bias current
- $\Delta\beta$ is normalized but V_{TH} isn't

Mismatches Dependence on Channel Area

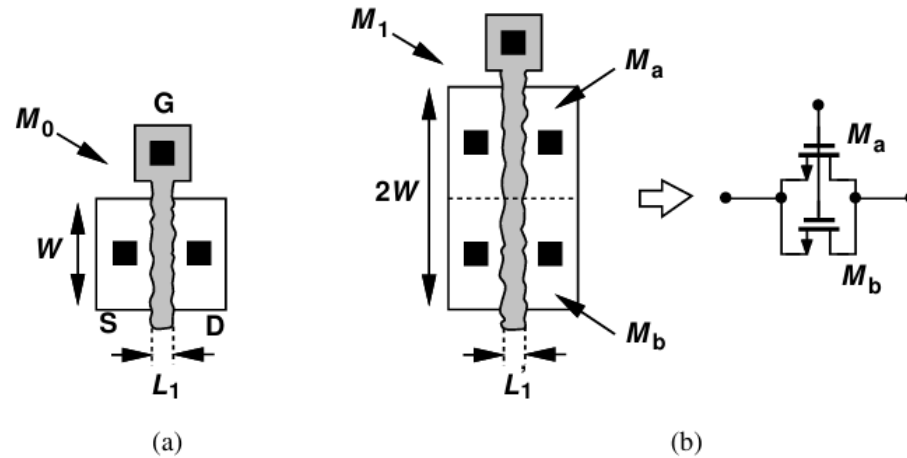


Figure 6.3 (a) Effect of edge “roughness,” and (b) reduction of mismatch due to a larger gate area.

$$\frac{W}{L_a} + \frac{W}{L_b} \approx \frac{2W}{L_0 + \frac{\delta L_a + \delta L_b}{2}}. \quad I_{D1} = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L_a} + \frac{W}{L_b} \right) (V_{GS} - V_{TH})^2,$$

$$I_{Da} + I_{Db} \approx \frac{1}{2} \mu_n C_{ox} \frac{2W}{L} \left[V_{GS} - V_{TH0} - \frac{\delta V_a + \delta V_b}{2} \right]^2.$$

$$V_{THa} = V_{TH0} + \delta V_a \quad V_{THb} = V_{TH0} + \delta V_b.$$

$$\frac{\Delta \beta}{\beta} = \frac{A_\beta}{\sqrt{WL}},$$

$$\Delta V_{TH} = \frac{A_{VTH}}{\sqrt{WL}}.$$

- σL & σV_{TH} falls by $\sqrt{2}$

Metastability

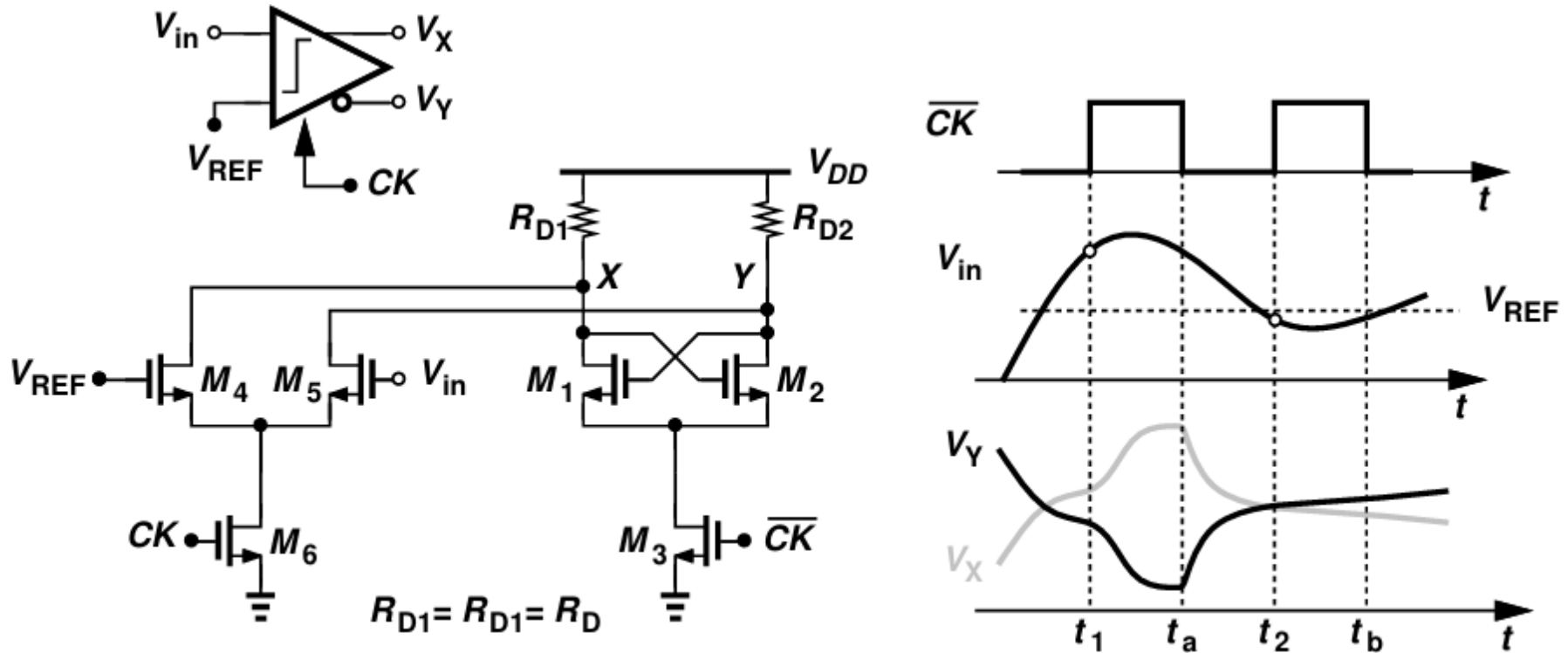


Figure 6.4 Metastability in a current-steering comparator.

- Metastability occur with a finite probability

Metastability: Regenerative Pair

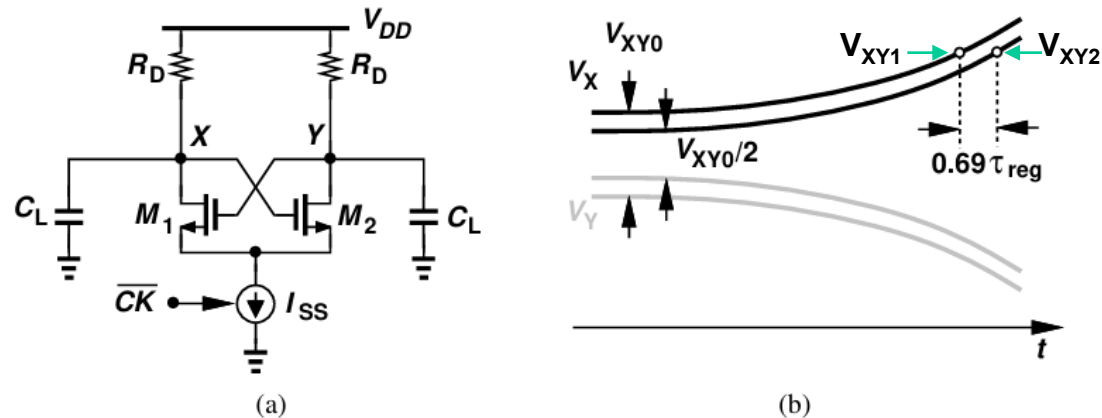


Figure 6.5 (a) Simple regenerative pair, and (b) metastability response for different initial conditions.

$$V_{XY1}(t) = V_{XY0} \exp \frac{t}{\tau_{reg}},$$

$$V_{XY2}(t) = (V_{XY0}/2) \exp \frac{t}{\tau_{reg}} = V_{XY0} \exp \frac{t - 0.69\tau_{reg}}{\tau_{reg}}.$$

- Halving initial difference, shifted the plot by $0.69\tau_{reg}$

Error Rate: Uniform Distribution

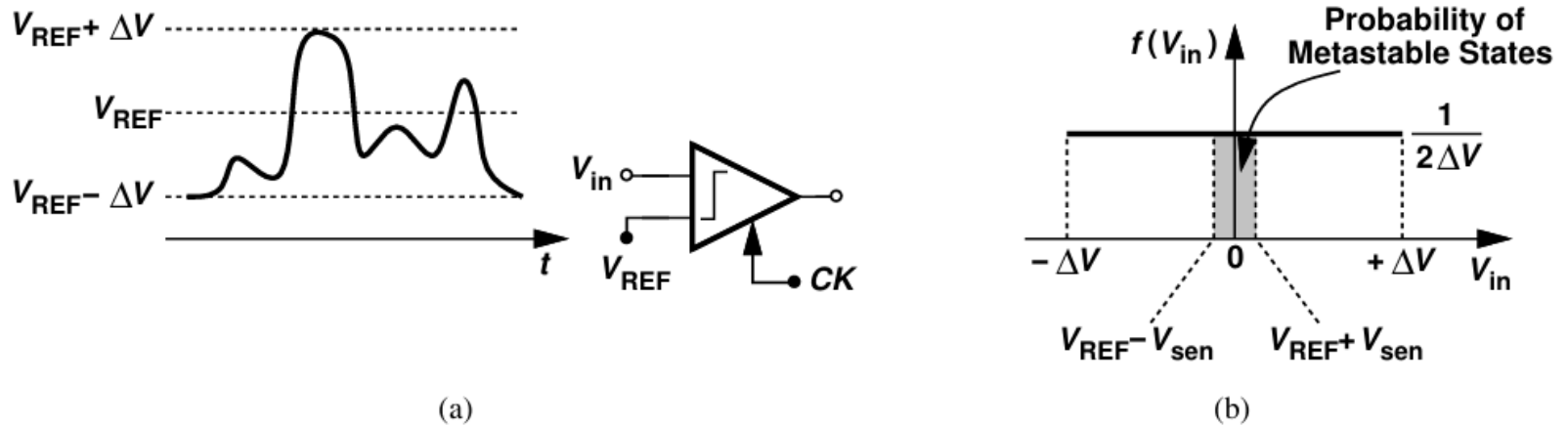


Figure 6.6 (a) Random signal swinging between two values, and (b) distribution of the signal.

$$|V_{in} - V_{REF}| \leq V_{sen} = (0.9I_{SS}/g_m) \exp(-T_m/\tau_{reg})$$

$$(2V_{sen})/(2\Delta V) = P_{error} = \frac{0.9I_{SS} \exp \frac{-T_m}{\tau_{reg}}}{g_m \Delta V}.$$

- This is assuming $f(V_{in})$ has a uniform distribution

Error rate: Gaussian Distribution

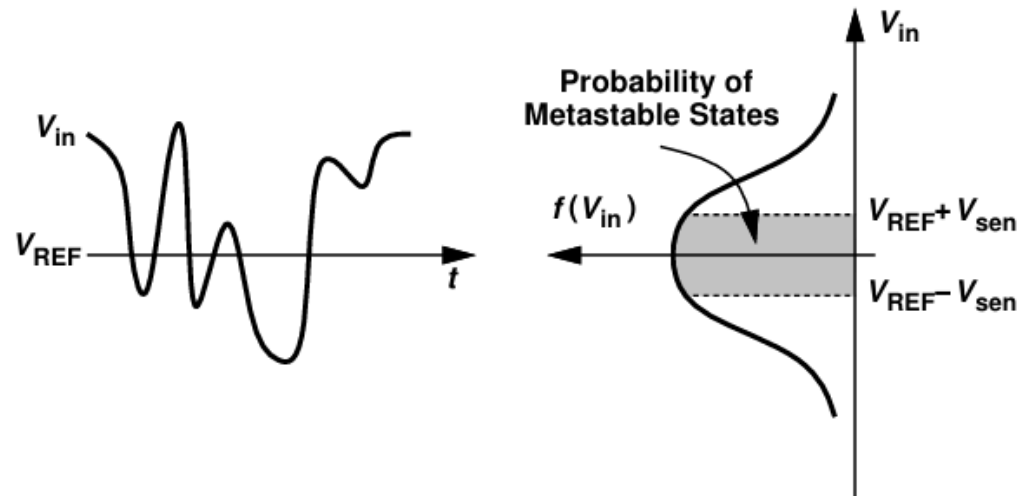


Figure 6.7 Random signal with Gaussian distribution.

$$f(V_{in}) = \frac{1}{\sqrt{2\pi}\sigma} \exp \frac{-(V_{in} - V_{REF})^2}{2\sigma^2}.$$

$$P_{\text{error}} = \frac{1}{\sqrt{2\pi}\sigma} \frac{1.8I_{SS}}{g_m} \exp \frac{-T_m}{\tau_{reg}}.$$

- **Max. swing isn't meaningful**
- **$P_{\text{error,uniform}} = P_{\text{error,gaussian}} \times (1/\Delta V)/[2/(\sqrt{2\pi}\sigma)]$**
- **Assuming $\Delta V = 3\sigma$, $P_{\text{error,uniform}} = 0.41 \times P_{\text{error,gaussian}}$**

Metastability: StrongArm Comparator

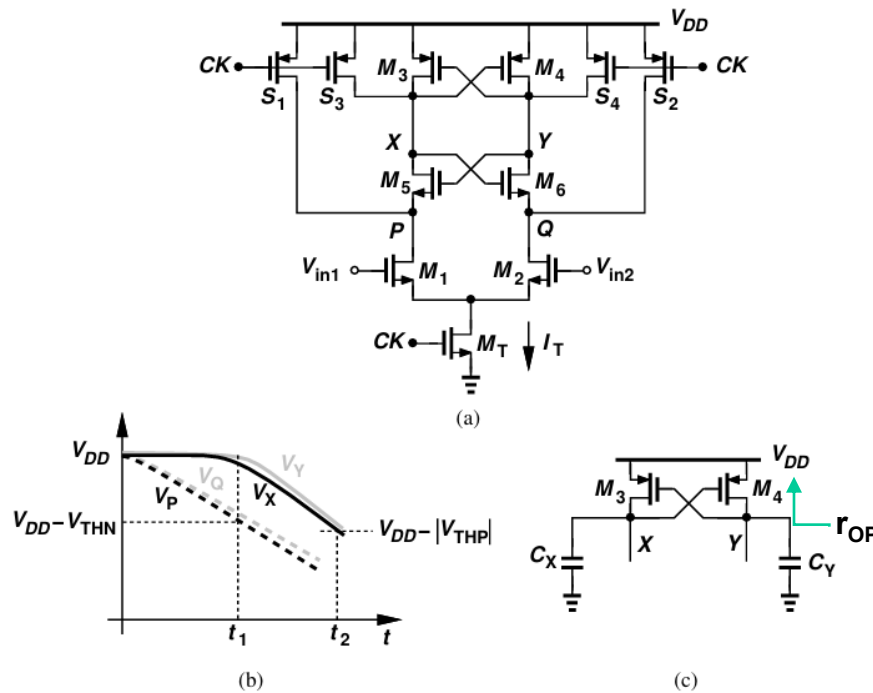


Figure 6.8 (a) StrongArm comparator, (b) its internal waveforms, and (c) PMOS pair in the regeneration mode.

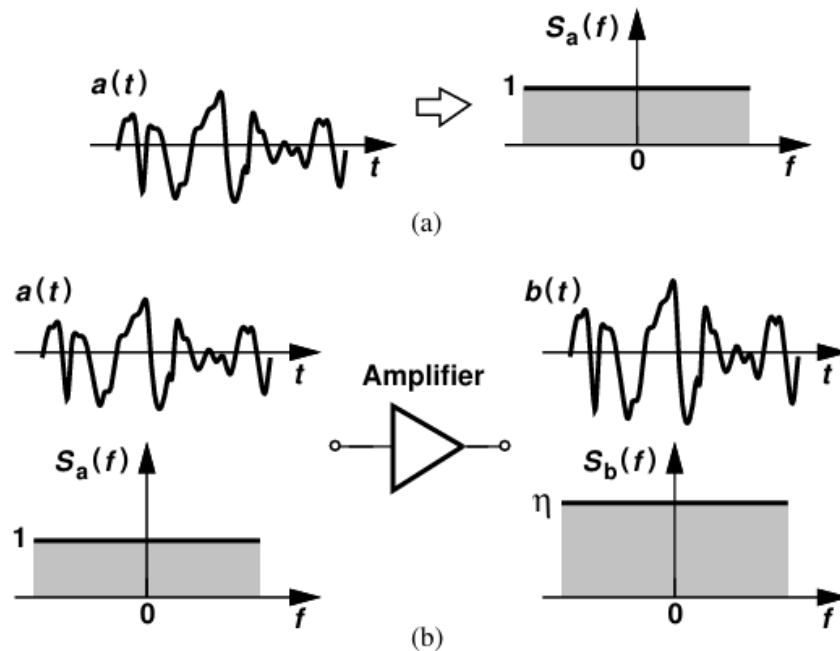
$$V_{XY} = V_{XY0} \exp \frac{t}{\tau_{reg}}$$

$$\tau_{reg} = \frac{r_{OP}}{g_{mP}r_{OP} - 1} C_L.$$

$$P_{error,uniform} = \frac{0.9V_{DD}}{A_0\Delta V} \exp \frac{-T_m}{\tau_{reg}}.$$

- V_{XY0} is the difference when M3 & M4 turns on
- $P_{error,uniform} \propto 1/A_0$

Noise in Time Domain



$$P_{avg} = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{+T/2} x^2(t) dt.$$

$$P_{avg} = E[x^2(t)],$$

Figure 6.9 (a) White noise in time and frequency domains, and (b) white noise traveling through an amplifier.

- Time domain noise multiplied by $\sqrt{\eta}$

Noise: Integrator

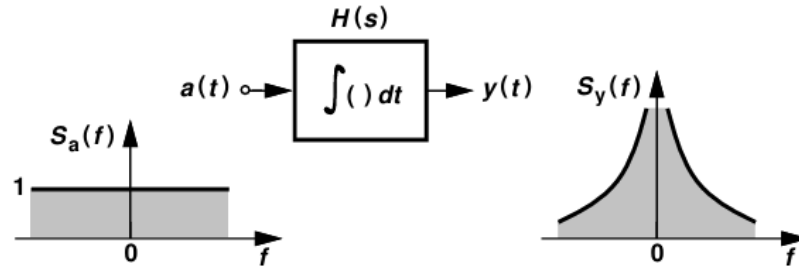


Figure 6.10 White noise applied to an integrator.

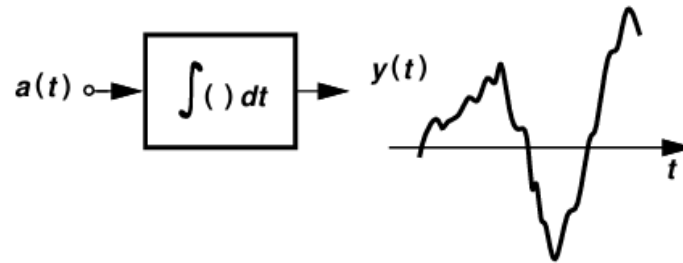


Figure 6.11 Growth of white noise at the output of an integrator.

$$E[y^2(t)] = E\left[\left(\int a(t)dt\right)^2\right] = E\left[\int dt\right] = t.$$

- Using Itô isometry, Integral of White noise is unbounded

White Noise Growth Example

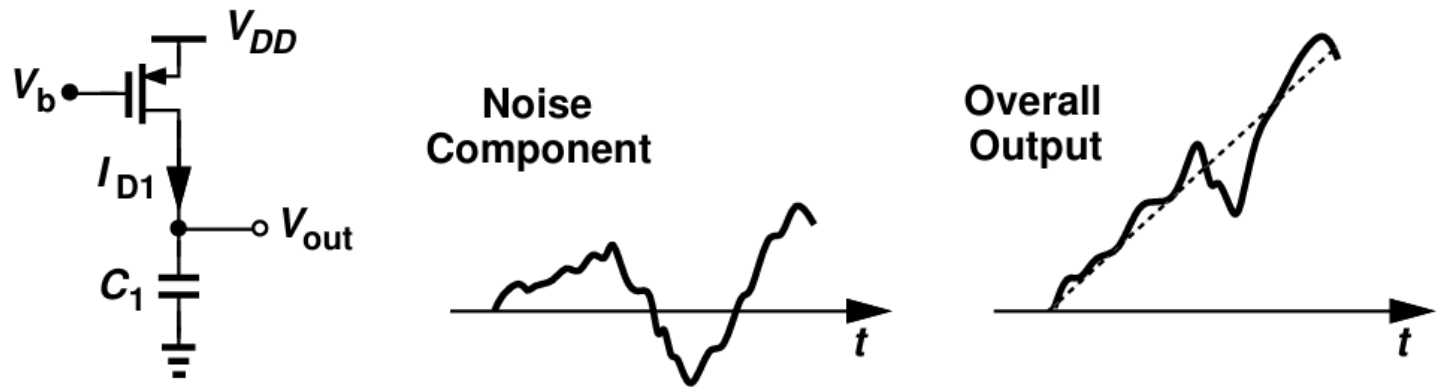


Figure 6.12 Example of white noise growth.

$$E \left[\left(\frac{1}{C_1} \int \sqrt{2kT\gamma g_m} a(t) dt \right)^2 \right] = \frac{2kT\gamma g_m}{C_1^2} t,$$

- Thermal noise component of (I_{D1}) is constant

Noise in Amplification Mode: StrongArm

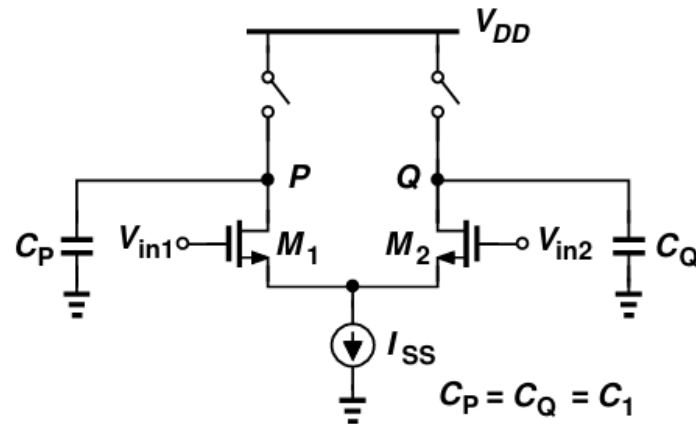


Figure 6.13 StrongArm comparator in its first amplification mode.

$$V_{n,in,rms} = \sqrt{\frac{2kT\gamma I_{SS}}{g_m C_1 V_{THN}} + \frac{kT}{C_1} \cdot \frac{I_{SS}^2}{2g_m^2 V_{THN}^2}}.$$

- A reasonable approximation from transistor thermal noise & kT/C noise.

Simulation of Comparator Noise

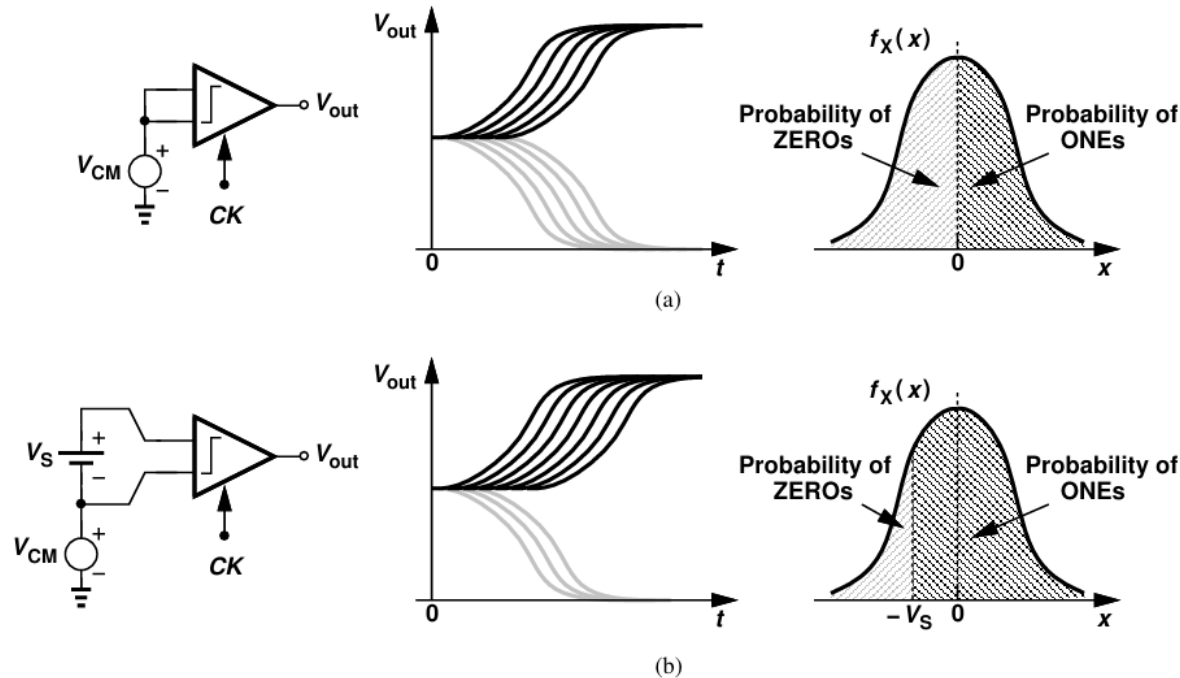


Figure 6.14 A comparator sensing a (a) zero input difference and its output behavior, and (b) a finite input difference and its output behavior .

$$\frac{\int_{-\infty}^{-V_S} \frac{1}{\sqrt{2\pi}\sigma} \exp \frac{-x^2}{2\sigma^2} dx}{1 - \int_{-\infty}^{-V_S} \frac{1}{\sqrt{2\pi}\sigma} \exp \frac{-x^2}{2\sigma^2} dx} = \frac{n_0}{n_1}.$$

- **Simulate comparator with no input (Gaussian noise source added), then simulate it with a small differential input ($V_S = \sim \text{mV}$)**

Example: Choosing $V_S = \sigma$

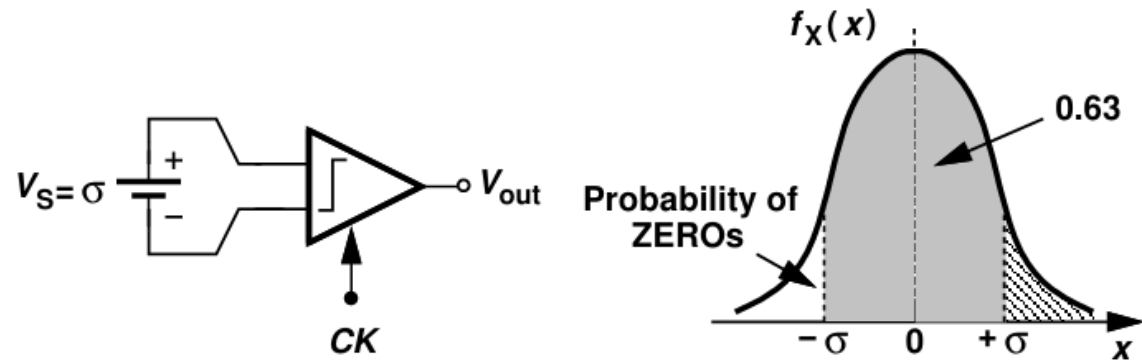


Figure 6.15 Behavior of a comparator with an input imbalance equal to its rms noise.

$\frac{n_0}{n_1} = \frac{0.185}{1 - 0.185}$ $= 0.227.$	$\frac{n_0}{n_1 + n_0} = \frac{0.227}{1 + 0.227}$ $= 0.185,$
--	--

- From simulation, adjust V_S to make it equal to around 0.185 value.

Offset Cancellation: Simple Diff. Pair

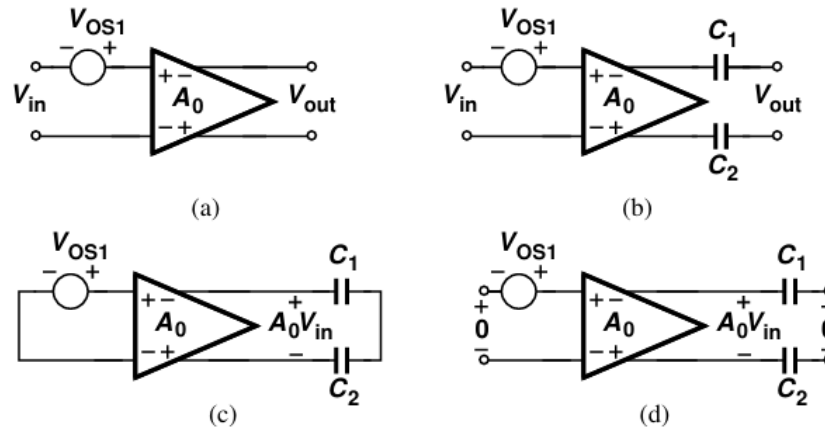


Figure 6.16 (a) An amplifier with an input-referred offset, (b) addition of capacitors at the output, (c) storage of output offset with a zero input, and (d) circuit after offset cancellation.

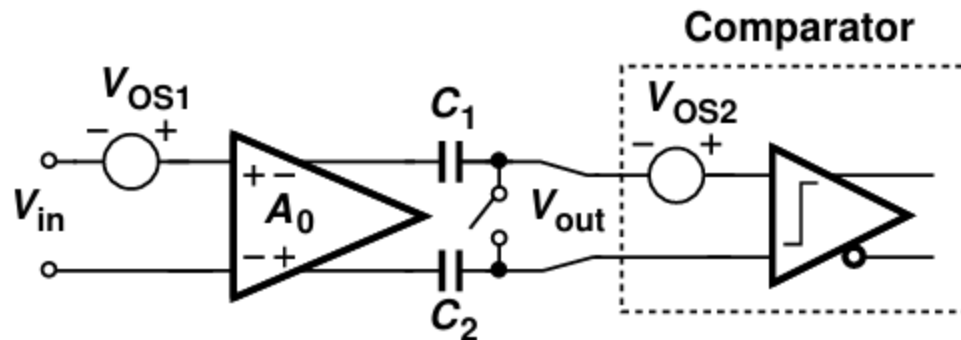


Figure 6.17 Offset-canceled amplifier followed by a comparator.

- **Offset stored in the output capacitors by shorting its output.**

Offset Cancellation Implementation

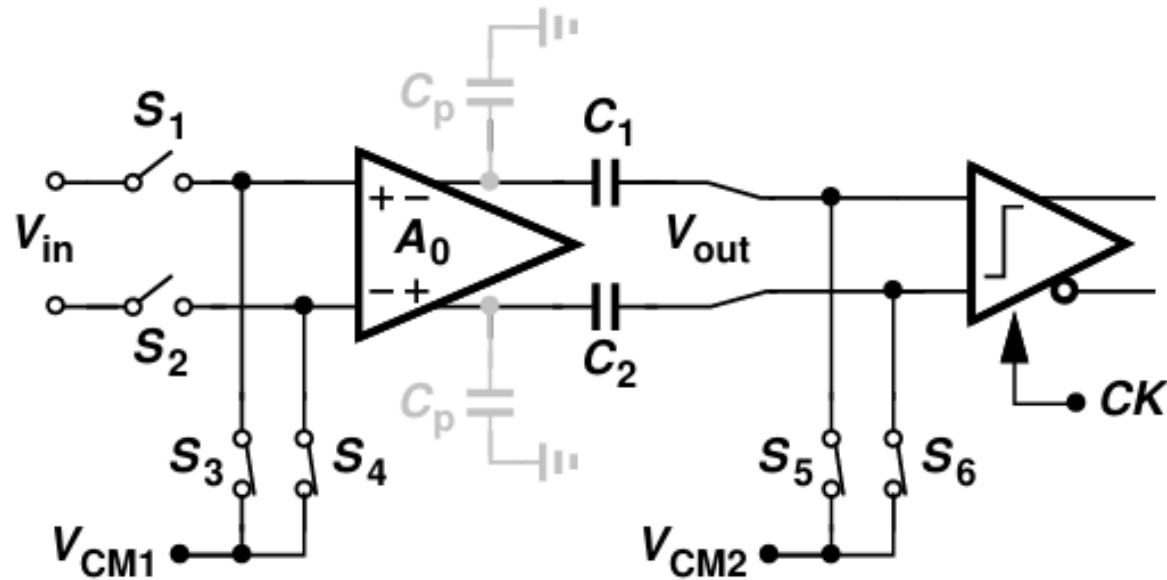


Figure 6.19 Detailed switching for offset cancellation.

$$\begin{aligned} V_{OS,in} &= V_{OS2} \div \left(\frac{C_1}{C_1 + 2C_{in}} A_0 \right) \\ &= \frac{V_{OS2}}{A_0} \left(1 + \frac{2C_{in}}{C_1} \right). \end{aligned}$$

- V_{CM1} & V_{CM2} for A_0 biasing
- Comparator offset reduced by $1/A_0$

Offset Cancellation at the input

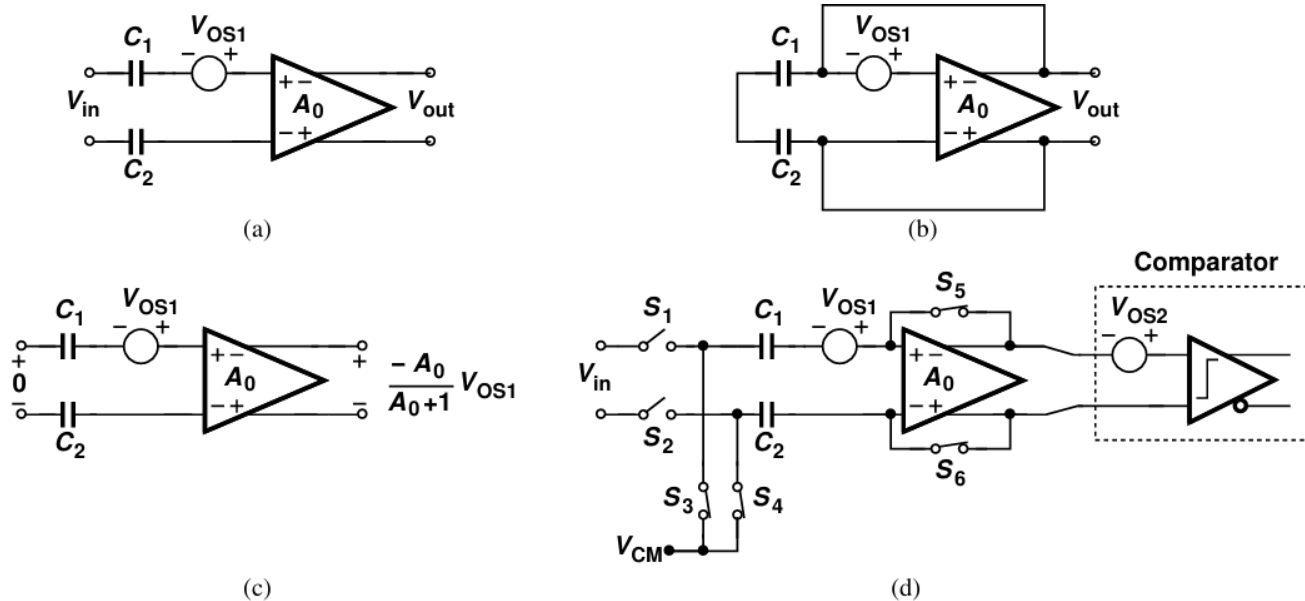
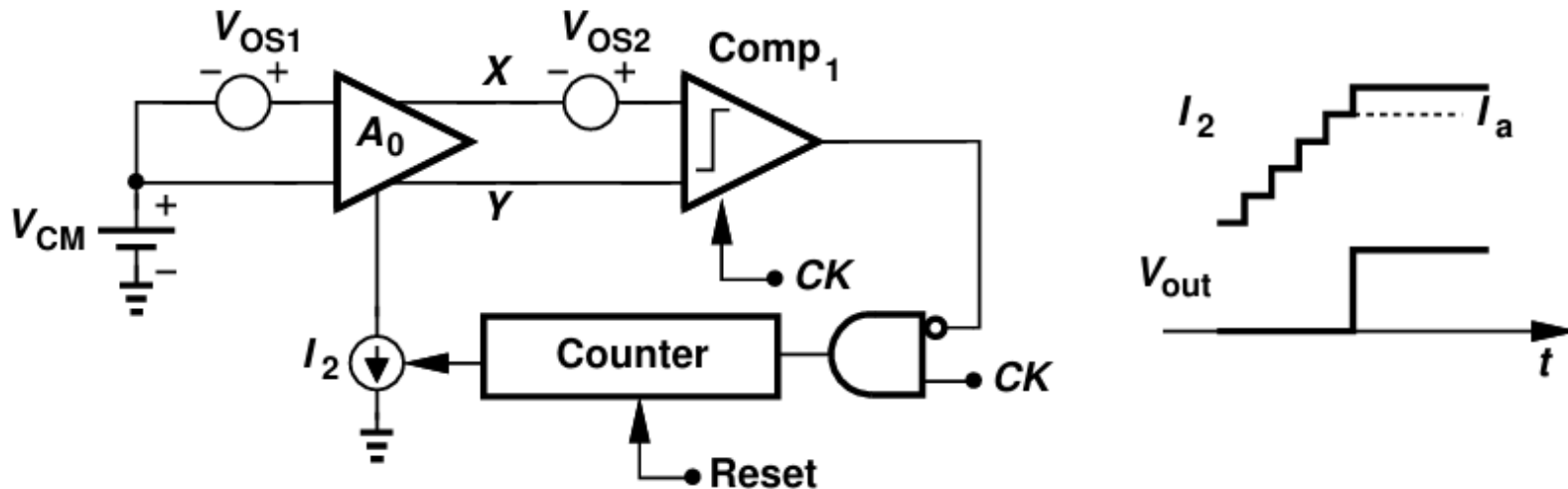


Figure 6.21 (a) Amplifier with input-offset storage capacitors, (b) offset cancellation mode, (c) residual offset at the output, and (d) amplifier followed by a comparator.

$$V_{OS,in} = \frac{V_{OS1}}{1 + A_0} \left(1 + \frac{2C_{in}}{C_1} \right) + \frac{V_{OS2}}{A_0} \left(1 + \frac{2C_{in}}{C_1} \right).$$

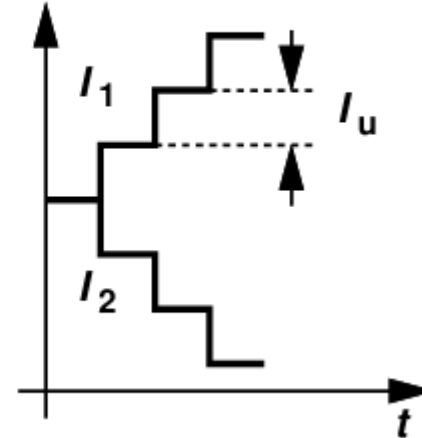
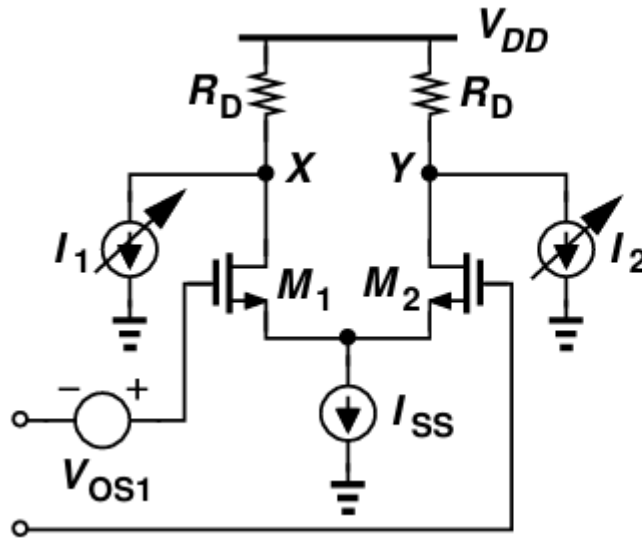
- V_{OS1} isn't cancelled completely
- A_0 has a gain ~ 3 or 4

Digital Offset Cancellation



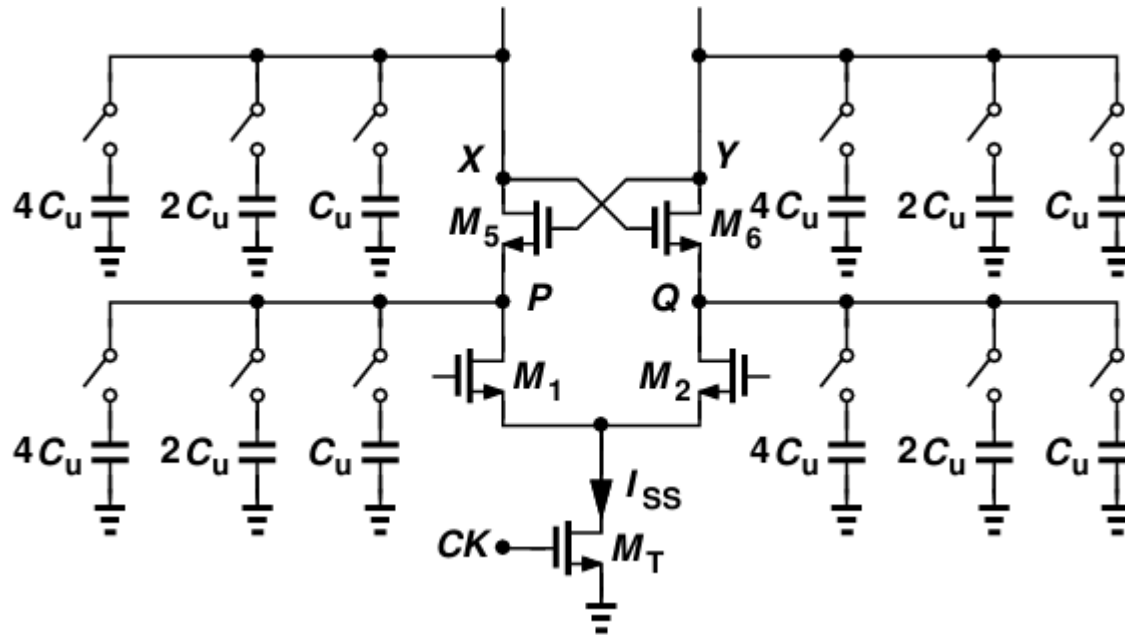
- Lowering node V_Y by drawing current from it.
- Consumes less Area than its analog counterparts
- A_0 & Comp_1 implemented as one comparator
- If $V_{OS1} < 0$,

Digital Offset Cancellation: $V_{OS1} < 0$



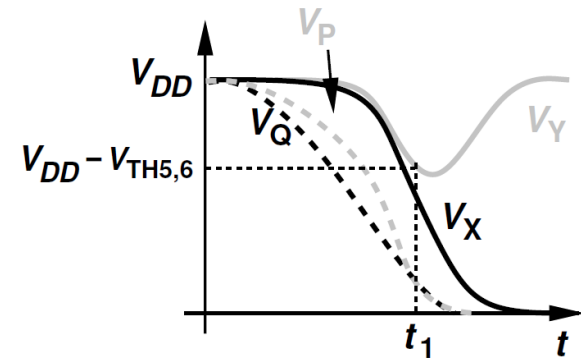
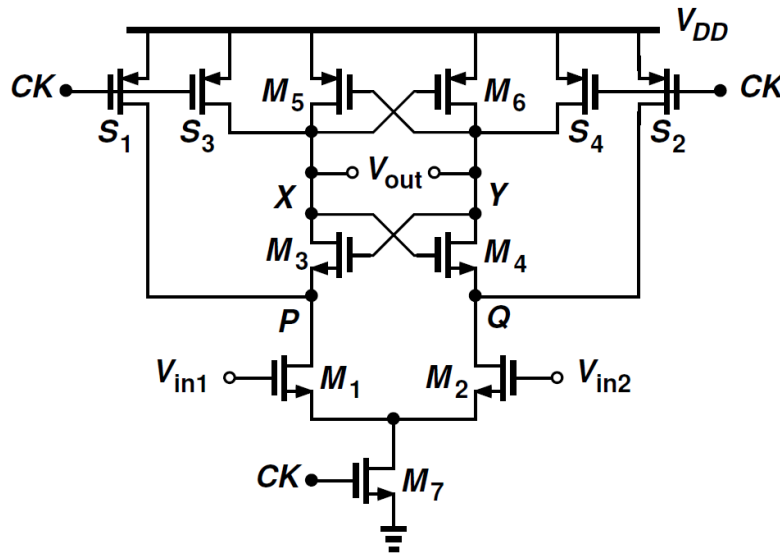
- Logic decides whether $I_1 - I_2$ increase or decrease at the beginning of the iteration.

Digital Offset Cancellation: StrongArm



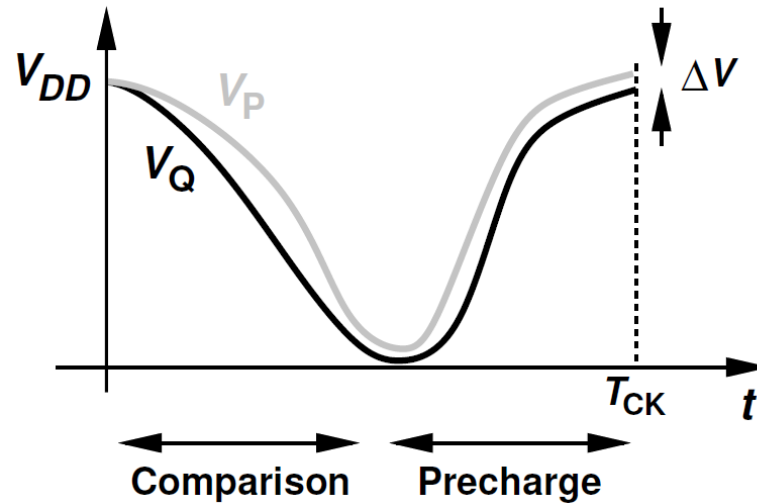
- capacitor arrays deliberately introduces mismatch.
- Degrades speed & burns more power.
- Large area from capacitor arrays.

Comparator Design Example



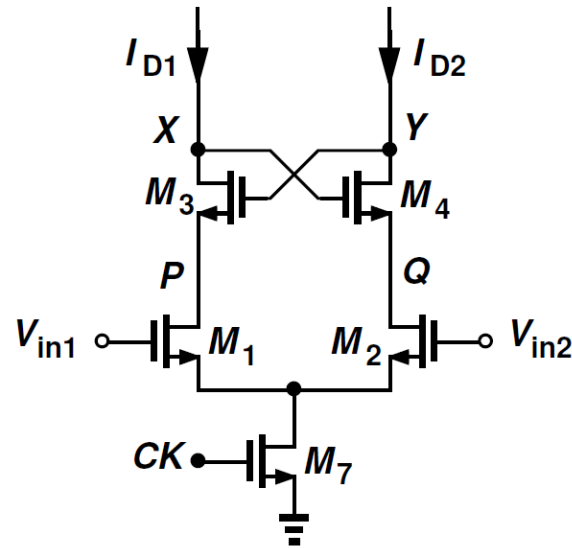
- Single clock phase, no static power, input offset primarily from the input pair, rail-to-rail output swings
- Precharge action:
- V_P and V_Q start from $V_{DD} \rightarrow M_1$ and M_2 in saturation
- Equalized states $\rightarrow$ suppressing “dynamic” offsets

General Considerations



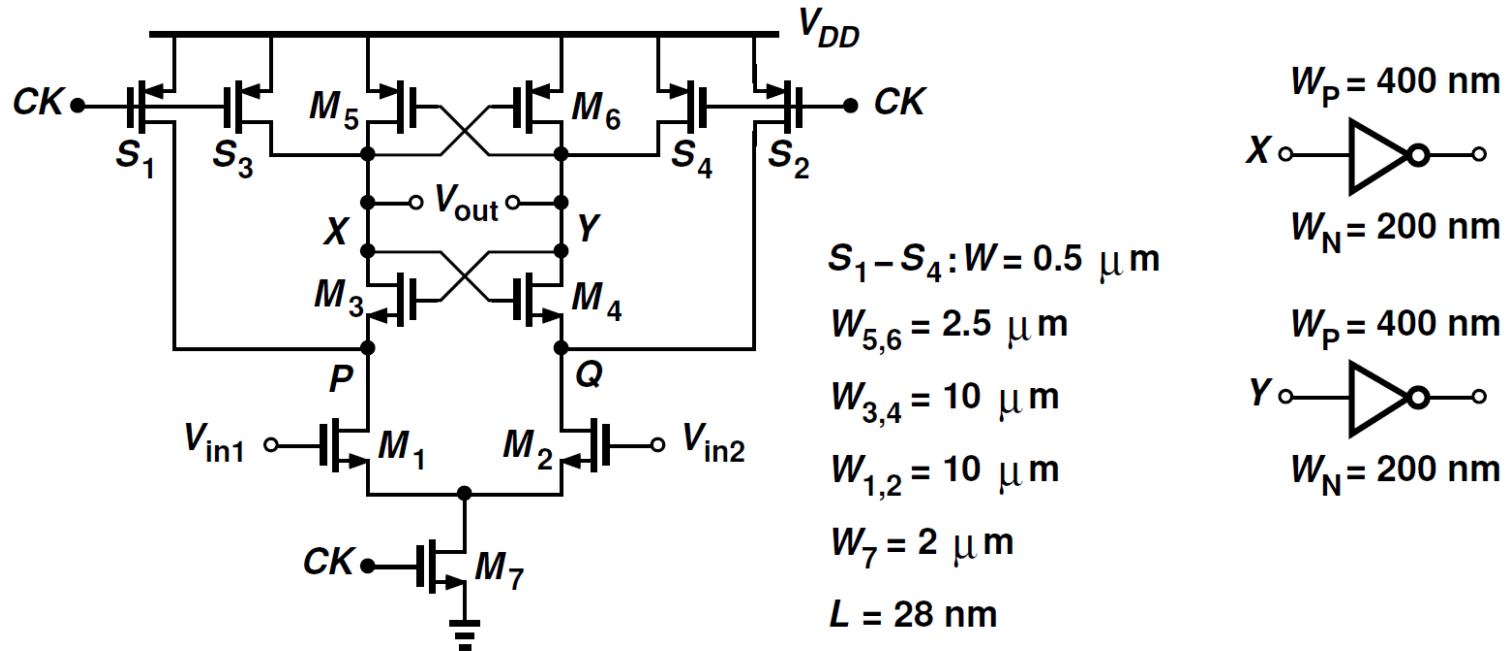
- ΔV at the end of precharge mode $\rightarrow$ amplification mode begins with ΔV stored on C_P and C_Q

Choice of Device Dimensions



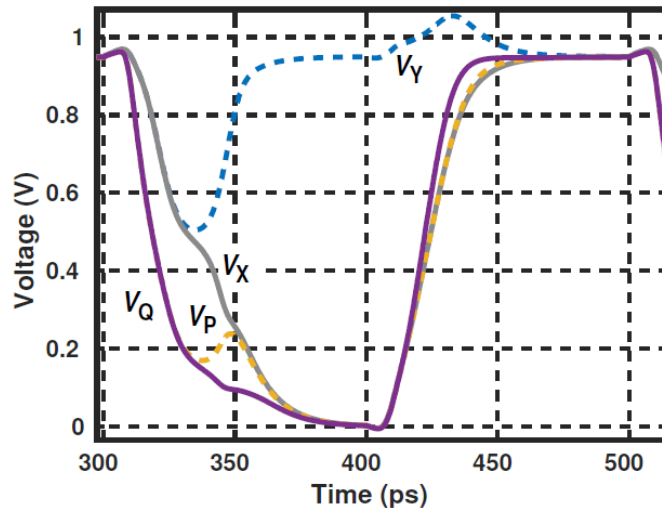
- **Worst case: SS corner, $V_{DD} = 0.95V$, $T = 75^\circ C$**
- $$\Delta V_{TH1,2} = \frac{A_{VTH}}{\sqrt{(WL)_{1,2}}} = \frac{2.2mV \cdot \mu m}{\sqrt{10\mu m \cdot 25nm}} = 4.4mV < 5mV$$
- $V_{GS7} = V_{DD}$, $V_{DS7} = 0.15V \rightarrow$ deep triode
- $W_7 = 2\mu m$, $W_{3,4} = 10\mu m$

Basic Waveforms

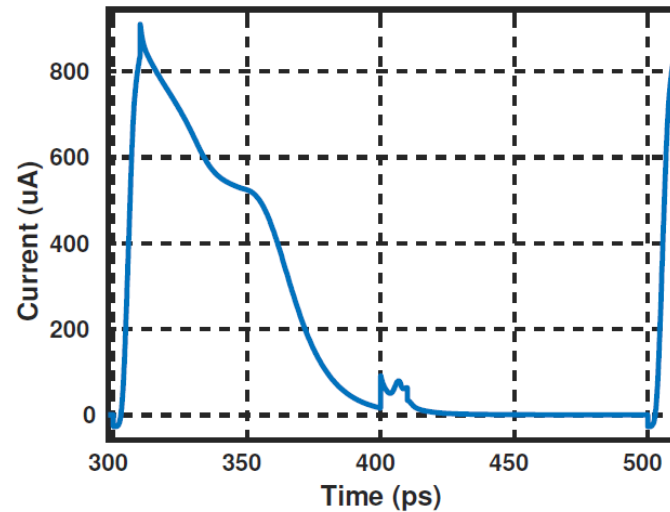


- output inverters $\rightarrow$ buffers
- Speed depends on input voltage difference.
- Apply a difference of 1 mV $\rightarrow$ “slow motion”

Basic Waveforms



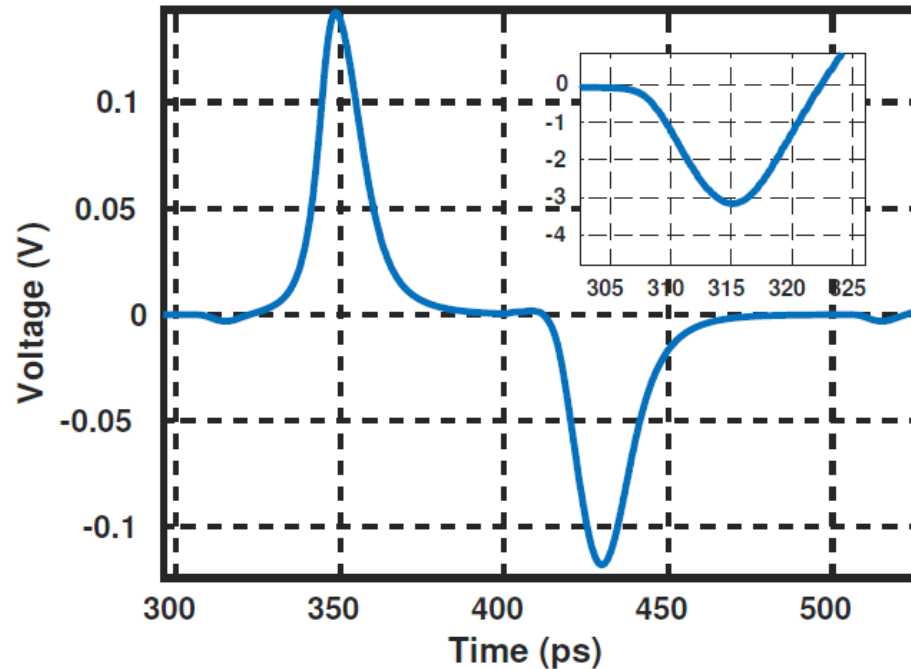
(a)



(b)

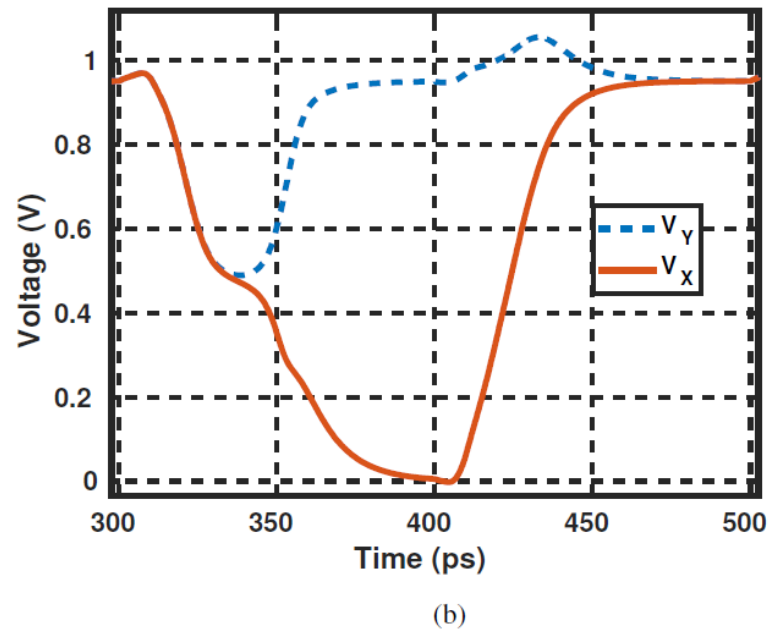
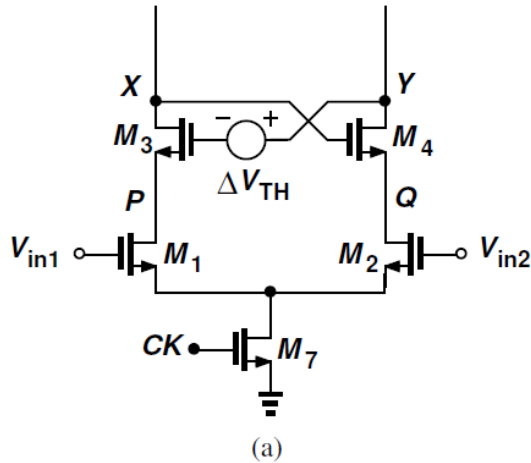
- V_X and V_Y experience a common-mode drop of about 400 mV before regeneration.
- Tail current reaches 800 μ A $\rightarrow$ input transistors triode and tail node voltage collapse

Basic Waveforms



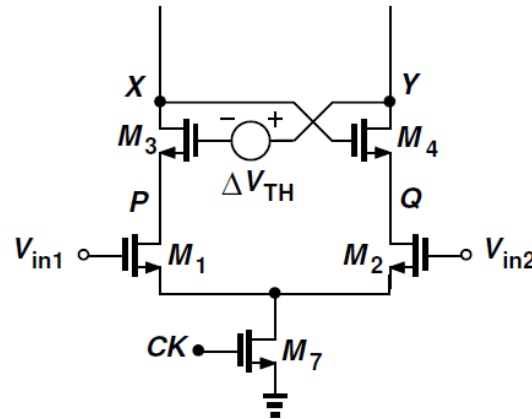
- **-3.15 mV @315 ps → initial voltage gain = 3.15**
- **Precharge devices strong enough and dynamic offset negligible**

Offset and Speed Optimization



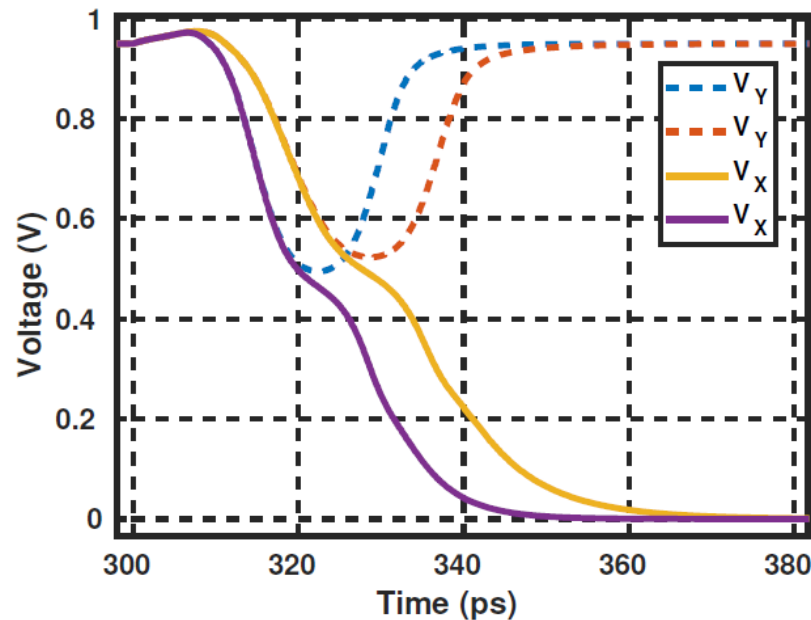
- $\Delta V_{TH3,4} = 4.4mV$ and find $V_{in1} - V_{in2} \approx 1.15mV$ nearly balanced
- Input referred offset of M_3 and M_4 : $4.4/1.15 = 3.8$
- Offset standard deviation: $\sqrt{(4.4mV)^2 + (1.15mV)^2} = 4.5mV$

Offset and Speed Optimization



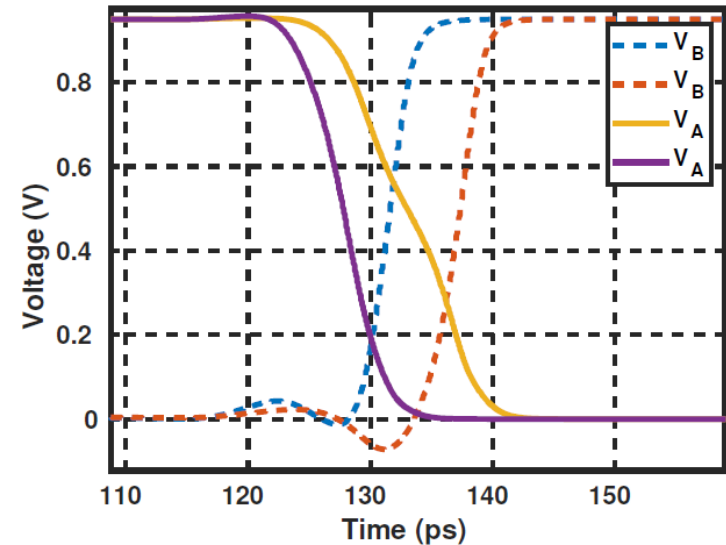
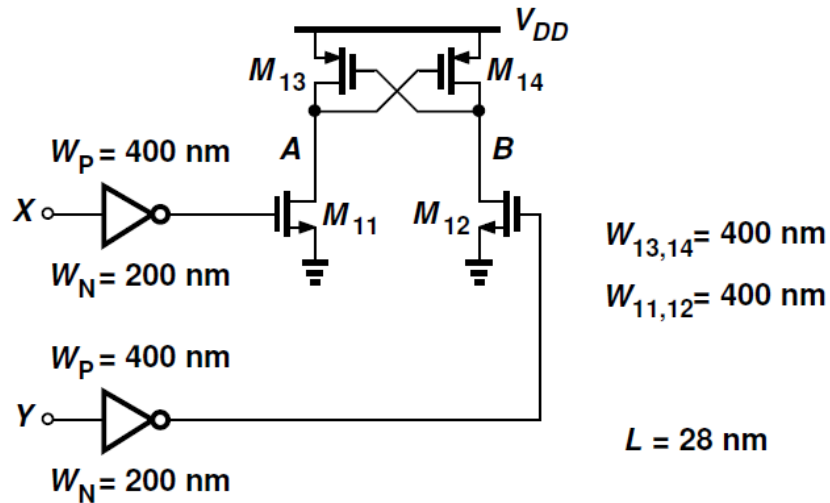
- **M_{3,4} Optimization:** $W_{3,4} = 5\mu\text{m}$, $\Delta V_{TH3,4} = \frac{A_{VTH}}{\sqrt{(WL)_{3,4}}} = 6.2\text{mV}$,
offset = $\sqrt{(4.4\text{mV})^2 + (1.5\text{mV})^2} = 4.6\text{mV}$ but speed increased
- **M_{5,6} Optimization:** $W_{3,4} = 2.5\mu\text{m}$, $\Delta V_{TH5,6} = \frac{A_{VTH}}{\sqrt{(WL)_{5,6}}} = 8.8\text{mV}$,
offset = $\sqrt{(4.4\text{mV})^2 + (1.5\text{mV})^2 + (0.9\text{mV})^2} = 4.7\text{mV}$
- Use Monte Carlo simulations to verify the calculations.

Offset and Speed Optimization



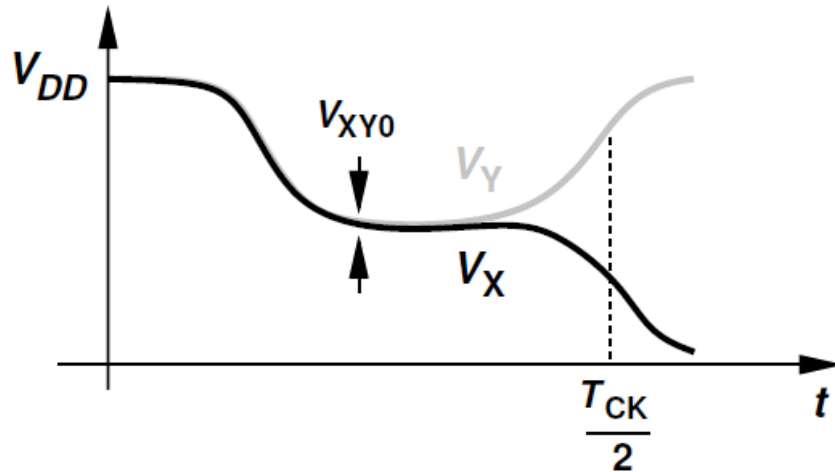
- Increase comparators' speed by widening M_7
- $W_7 = 2\mu\text{m}$ and $4\mu\text{m} \rightarrow 28\text{ps}$ and 22ps
- Signal path power consumption: $2f_{CK}C_PV_{DD}^2 + f_{CK}C_XV_{DD}^2$ constant
- Clock path draws greater power.

Addition of RS Latch

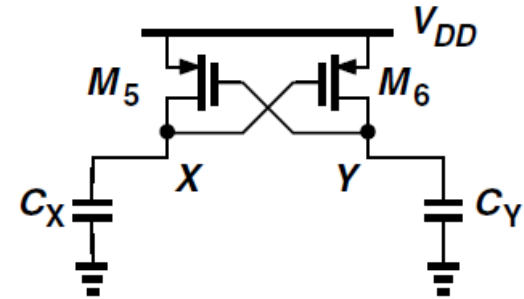


- To present a valid logical level in precharge mode
- Minimize delay $\rightarrow$ wide $W_{13,14}$, wide $W_{11,12}$ and wide W_P
- $W_{11,12} = W_P = 800 \text{ nm} \rightarrow 6 \text{ ps}$ delay drop

Metastability



(a)



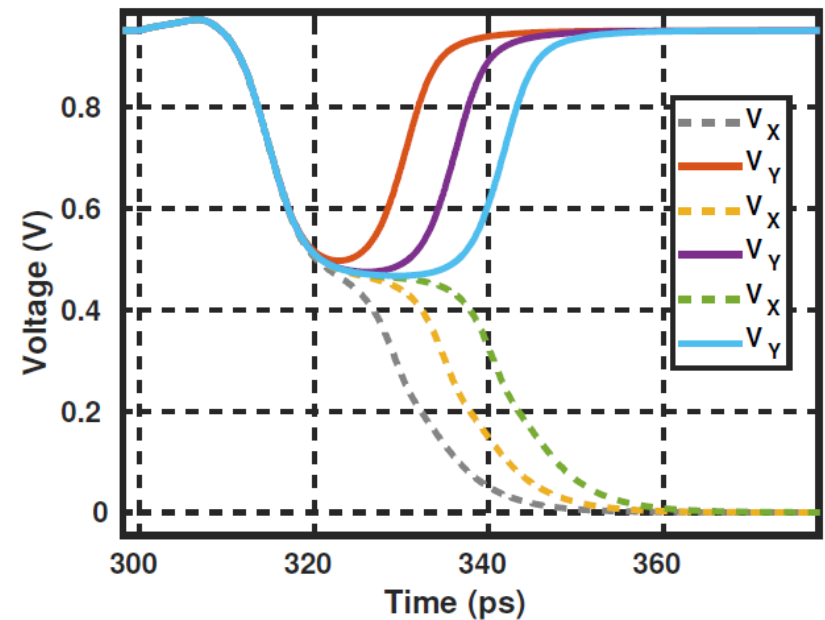
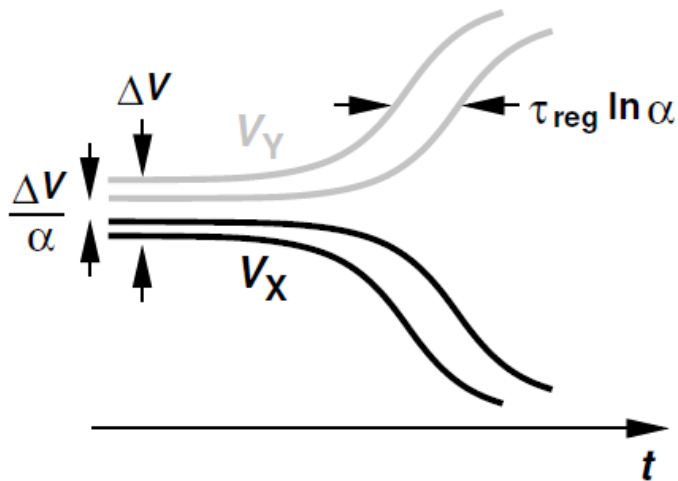
(b)

- **Positive feedback:** $V_{XY} = V_{XY0} \exp \frac{t}{\tau_{reg}}$, $\tau_{ref} = \frac{g_{m5,6}}{C_X}$

- **Minimum acceptable value V_1 :**

$$V_{XY0} \geq V_1 \exp \frac{-T_{CK}/2}{\tau_{reg}}$$

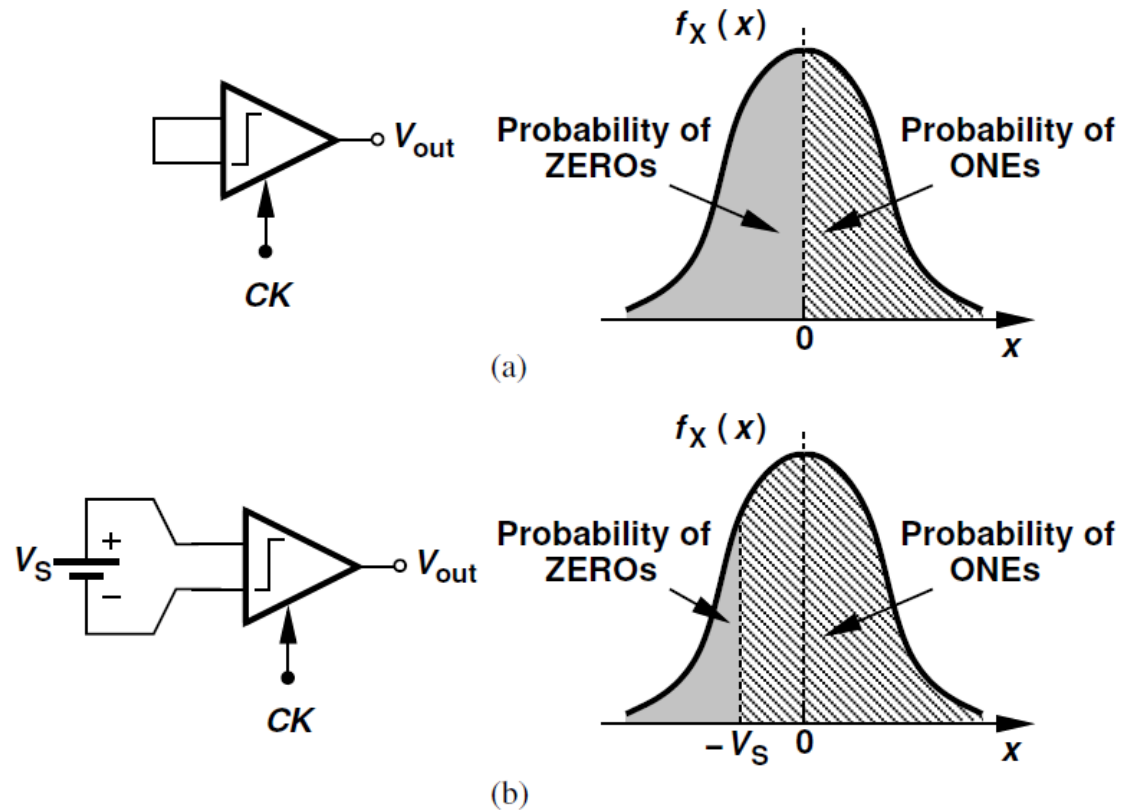
Metastability



- $$V_{XY2} = \frac{\Delta V}{\alpha} \exp \frac{t}{\tau_{reg}} = \Delta V \exp(-\ln \alpha) \exp \frac{t}{\tau_{reg}}$$

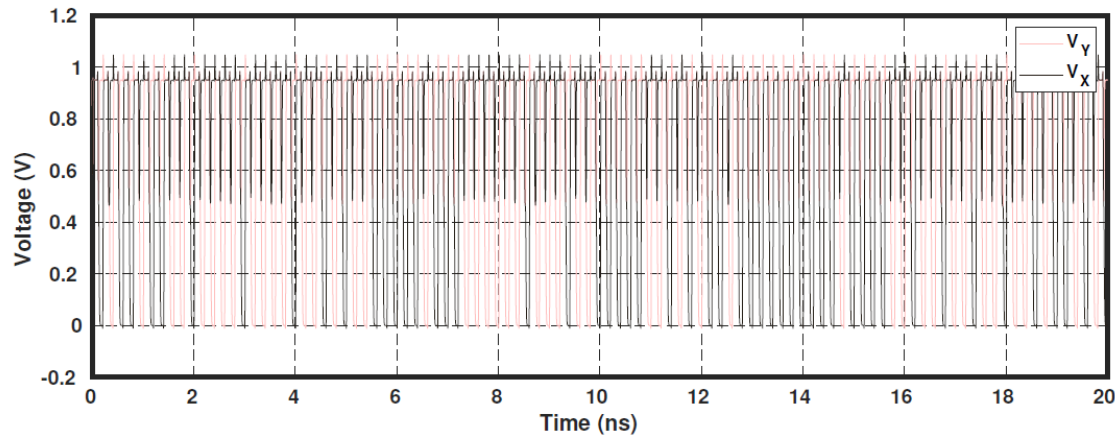
$$= \Delta V \exp \frac{t - \tau_{reg} \ln \alpha}{\tau_{reg}}$$
- $$1\text{mV}, 100\mu\text{V}, 10\mu\text{V} \rightarrow \tau_{reg} = \frac{5.7\text{ps}}{\ln 10} = 2.5\text{ps}$$

Input-Referred Noise

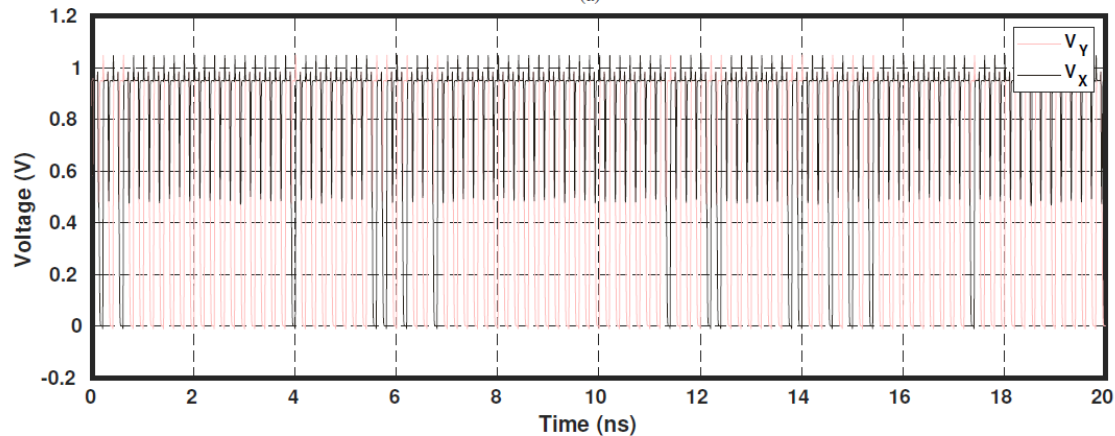


- Digital outputs $\rightarrow$ transient noise simulation
- Without $V_S \rightarrow$ 0s and 1s with equal probabilities
- With $V_S \rightarrow$ probabilities of 0s reduced

Input-Referred Noise



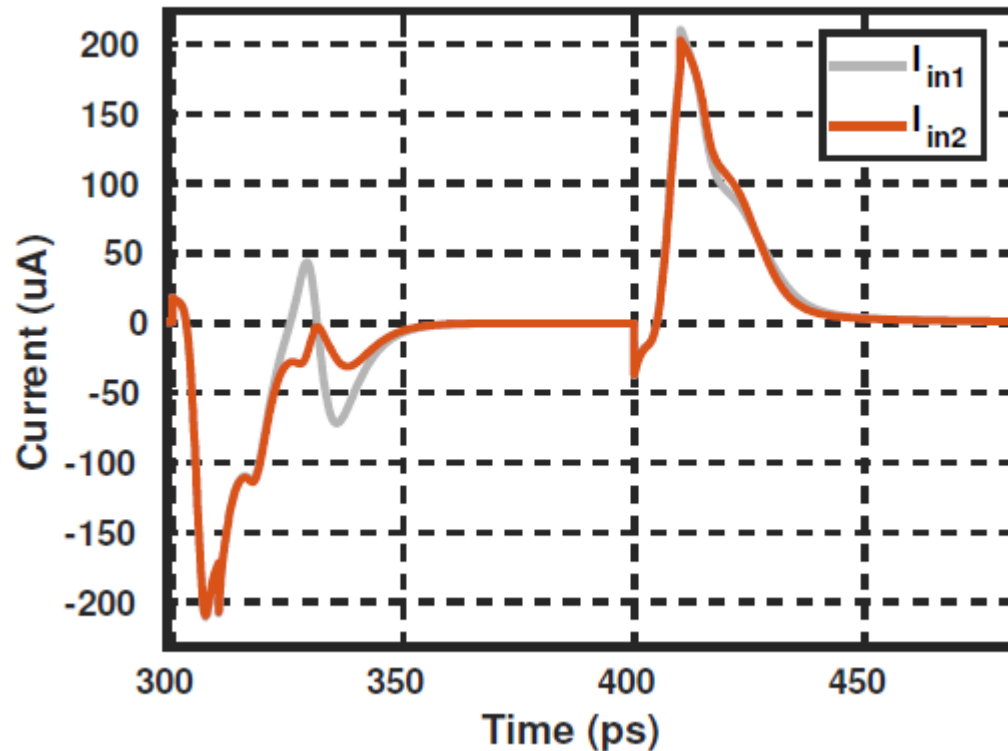
(a)



(b)

- Gaussian distribution from $-\infty$ to $-\sigma$: 16%
- (a) V_X goes to zero 50 times
- (a) V_X goes to zero 16 times, $V_S = \sigma$

Kickback Noise



- Large transient currents from inputs $\rightarrow$ kickback noise
- common-mode components $\rightarrow$ objectionable
- differential components $\rightarrow$ problematic generally